Representation, Topology, and a Blood–Brain-Barrier-Associated Slow-Wave Marker: A Controlled Graph Neural Network Study of EEG in Alzheimer's Disease and Frontotemporal Dementia

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Draft — October 11, 2026

Abstract

Objective. EEG-based classification of Alzheimer's disease (AD) and frontotemporal dementia (FTD) on the same public dataset reports accuracies from 80% to 99%, but studies change the signal representation, graph construction, model and evaluation protocol at once. We isolated these factors and tested a blood–brain-barrier (BBB)-associated EEG marker. *Approach.* Using OpenNeuro ds004504 (36 AD, 23 FTD, 29 controls), we crossed seven node-feature representations with three graph constructions under one fixed two-layer graph convolutional network, evaluated on 25 frozen subject-level cross-validation splits (525 runs) and re-evaluated on 25 independent confirmation splits, with a by-subject mixed model as the primary analysis, matched non-graph baselines, 16 ablations, label permutations, a leakage experiment, stopping-rule sensitivity, and external validation of the whole protocol on a Parkinson's disease cohort (ds004584, 149 participants). We also characterised paroxysmal slow-wave events (PSWEs), an EEG marker linked to BBB dysfunction. *Main results.* Representation dominated ($F(6, 510) = 16.60$, replicated at 19.81 on independent splits and 20.03 externally); graph construction mattered little, with graph types differing by less than 0.01 in the probability assigned to the true class, in a direction that differed between cohorts. The best configuration (wavelet features, spatial graph) reached macro-F1 0.596 on the confirmation splits (95% CI 0.554–0.643; permutation $p = 0.010$), not significantly better than non-graph classifiers; window-level splitting of the identical pipeline raised it to 0.983. Externally the same configuration reached 0.688, leakage again inflated it (0.839), but the ranking of representations did not transfer ($\rho = 0.29$). Fixed-threshold PSWEs were more frequent in AD (rate ratio 2.96) and FTD (2.07) but not in Parkinson's disease (1.21); in both cohorts the rate tracked background slowing ($\rho = 0.88$ and 0.81), robustly to spectral zero-padding and artefact rejection. *Significance.* The evaluation protocol and the signal representation, not the graph or the model, determine reported performance; representation rankings are disease-specific; and an EEG marker of BBB dysfunction must be defined relative to background slowing. Code, frozen folds and all raw runs are released.

Keywords: EEG; graph neural network; Alzheimer's disease; frontotemporal dementia; signal representation; functional connectivity; blood–brain barrier; paroxysmal slow-wave events; data leakage; reproducibility

1 Introduction

Alzheimer's disease (AD) and frontotemporal dementia (FTD) are leading causes of neurodegenerative dementia, particularly with onset before 65, and telling them apart is clinically consequential: they differ in prognosis, in care needs and in which disease-modifying treatments are appropriate. Scalp electroencephalography (EEG) is attractive for this task because it is cheap, non-invasive and available in routine care, in contrast to amyloid PET or MRI. Its physiological basis is well established: AD is increasingly understood as a disorder of large-scale brain networks [1], and graph-theoretical EEG studies have shown that AD and FTD disturb those networks in *different* ways [2], which is what makes a three-class AD/FTD/control problem scientifically meaningful rather than merely harder.

Machine-learning work on this question has moved from spectral markers and handcrafted features with classical classifiers [3] to convolutional networks and transformers [4, 5] and, most recently, to graph neural networks (GNNs) that treat electrodes as nodes and inter-electrode coupling as edges [6–12]. The public OpenNeuro dataset ds004504 [13, 14], with 88 subjects recorded on one device under one protocol, has become the de-facto benchmark. Reported accuracies on this single dataset span roughly 80–99%.

That spread is implausibly wide for identical data, and it hides a methodological problem. Published studies change several things at once: (i) the *signal representation* fed to the model, (ii) the *graph construction* that defines which electrodes are connected, (iii) the *model architecture*, and (iv) the *evaluation protocol*. Independent evidence suggests that the last of these matters most. A 2026 scoping review of this exact benchmark concluded that apparent deep-learning gains "arise not from inherent representational advantage, but from data leakage" [15], echoing a broader warning about leakage across machine-learning-based science [16] and in neuroimaging prediction specifically [17]. In clinical machine learning, splitting records rather than subjects has long been known to overestimate accuracy for new patients [18], and segment-level splitting is common in, and inflates, deep-learning studies of translational EEG [19, 20]. A 2026 comparison of five time–frequency representations under a single convolutional network [21] isolated representation, but its authors note that it "does not explicitly model the true spatial topology" and that graph-based schemes remain unexplored; others observe that most studies "overlook spatial connectivity between channels" [10], and explicitly call for graph convolutional networks in future work [22, 23]. No study has varied representation and graph construction together while holding the model and the evaluation protocol fixed.

A second, biologically motivated gap concerns the blood–brain barrier (BBB). BBB breakdown is an early biomarker of cognitive dysfunction [24, 25], and Milikovsky et al. described *paroxysmal slow-wave events* (PSWEs) — episodes in which the EEG median power frequency falls below 6 Hz for at least 5 s — in patients with AD, linked them to cognitive impairment, localised them to cortex with BBB dysfunction in epilepsy, and induced them in rodents by albumin extravasation [26, 27]. PSWEs have since been examined in Parkinson's disease [28] and epilepsy [29], but, to our knowledge, never in FTD and never on an open dementia cohort. Whether this BBB-associated EEG marker carries diagnostic information beyond ordinary EEG slowing has not been tested.

We therefore ran a controlled factorial experiment. Seven node-feature representations (spectral, connectivity, wavelet, time-domain, raw, fused, and PSWE-derived) were crossed with three graph constructions (spatial, functional, hybrid) under a single fixed two-layer graph convolutional network [30], with subject-level repeated cross-validation, identical frozen folds for every condition, and hyperparameters fixed before any result was seen. Because the model is the control variable rather than the contribution, it is deliberately small. Around this core we measured the inflation produced by window-level (leaky) splitting on the identical pipeline, quantified age/sex confounding, and characterised PSWE burden across AD, FTD and controls.

Our contributions are:

1. **Methodological.** A fully crossed 7×3 protocol that separates signal representation, graph topology and model architecture, with a by-subject mixed model as the primary analysis, paired statistics over 25 frozen subject-level test splits, an independent set of confirmation splits that removes selection bias, and ablations that test whether the graph matters at all.

2. **Empirical.** Effect sizes and confidence intervals for which representations a GNN benefits from in three-class AD/FTD/control classification, with non-graph twins and a demographic confound floor as reference points.

3. **Protocol.** A direct measurement, on one pipeline, of how much window-level splitting inflates results — a calibration factor for reading published numbers on this benchmark — and an external test of the whole protocol on a Parkinson's disease cohort.

4. **Biological.** The first characterisation of BBB-associated PSWEs in FTD, an independent-cohort test of the AD finding, a replication in Parkinson's disease, and a test of whether the marker is separable from background slowing.

5. **Reproducibility.** Code, seeds, frozen folds and every raw run released; the full study re-runs on free cloud CPUs.

2 Related work

Signal representations for EEG dementia classification. Spectral representations dominate: relative band power and EEG slowing are the most replicated AD markers [4, 31, 32]. Time-domain and complexity features (Hjorth parameters, entropies) are cheap and interpretable [5, 32]. Time–frequency transforms such as the continuous or discrete wavelet transform retain non-stationary structure that a power spectrum averages away [21, 33], and several recent studies encode EEG as images for convolutional networks [11, 21, 34]. Representation choice is thus treated as a first-class question — but almost always with convolutional models, and without varying the spatial structure the model sees.

Graph neural networks for EEG. Spectral graph convolution [30] and attention-based variants [35] underpin most EEG GNNs, reviewed in [6]. For dementia, GNNs have been built on functional-connectivity graphs [7], learned adjacency [8], multi-connectivity inputs [12], time-resolved graphs [11] and spatial–functional hybrids [10]; for Parkinson's disease, interpretable graph learning has been used to localise abnormal connectivity [23]. The closest precedent [7] compared connectivity measures as GNN inputs but held node features fixed. A 2025 preprint applied a graph transformer to ds004504 for FTD versus controls with window-level prediction [36]. None of these studies crosses node-feature representation with edge definition under a fixed model.

Graph construction for EEG. Edges may be defined spatially (inter-electrode distance), functionally (statistical coupling), by learning, or by combinations of these. Phase-lag-based measures [37, 38] suppress the zero-lag coupling produced by volume conduction, which otherwise inflates functional graphs; weighted graphs are recommended over binarised ones for EEG [2], and phase-synchronisation networks differ between dementia subtypes [33, 39]. Spatial graphs carry no subject-specific information on their own and act as a smoothing prior; whether that prior helps is an empirical question we test directly.

Table 1: Positioning against the closest prior work on EEG dementia classification. ✓ = yes; – = no; ? = not yet verified against the full text.

Study	Varies repr.	Varies graph	Fixed model	Subject-level split	Public data
DICE-Net [4]	–	–	✓	✓	✓
FC-measure GNN [7]	–	✓	✓	?	?
Adaptive GCN [8]	–	✓	–	?	?
TFR comparison [21]	✓	–	✓	?	✓
Spatial–functional GNN [10]	–	–	–	?	?
Multi-connectivity GCN [12]	–	✓	–	?	?
Graph transformer [36]	–	–	✓	✓	✓
This study	✓	✓	✓	✓	✓

Evaluation protocol and leakage. Leakage — information from test data reaching training — is a general failure mode of machine-learning-based science [16], and has been shown to inflate prediction performance in connectome-based neuroimaging models [17]. Saeb et al. showed that when the use case is prediction for new individuals, record-wise cross-validation, which lets one person's records appear in both training and test sets, greatly overestimates accuracy relative to subject-wise cross-validation [18]. In EEG the problem takes a specific form: windows cut from one recording are highly correlated, so splitting *windows* rather than *subjects* lets a model recognise individuals instead of disease. Individual differences and non-stationarity make the partitioning decision consequential for cross-participant EEG models, whose error rates improper partitioning can understate severalfold [40]. Brookshire et al. found that segment-based holdout strongly overestimated performance on unseen subjects in an Alzheimer's and an epilepsy dataset, and that most translational deep-learning EEG studies they surveyed used it [19]; a recent audit across ADHD and Parkinson's cohorts — the latter the same cohort we use for external validation — reproduced the inflation in both [20]. In connectome-based neuroimaging models, leakage through repeated subjects drastically inflated prediction performance, and more so in small datasets [17]. The 2026 scoping review of ds004504 attributes the upper end of reported accuracies on this benchmark to the same effect [15]. Best-practice guidance for prediction studies [41] and the large error bars of cross-validation in small samples [42] add a separate caution for cohorts of our size. We measure the leakage effect directly on one pipeline in both cohorts, and report subject-level confidence intervals throughout.

EEG and the blood–brain barrier. BBB breakdown precedes and predicts cognitive decline [24, 25]. Milikovsky et al. proposed PSWEs as an EEG manifestation of BBB-related cortical dysfunction in AD and epilepsy [26], with a companion study tracing the mechanism to albumin-activated TGFβ signalling [27]. Subsequent work found PSWEs uncommon in Parkinson's disease [28] and explored them as an epilepsy biomarker [29]. Because ds004504 includes no BBB imaging, we treat PSWEs strictly as a *BBB-dysfunction-associated* EEG marker, not as a measurement of BBB integrity.

Positioning. Table 1 summarises how this study differs from the closest prior work.

3 Materials and methods

Figure 1 summarises the design.

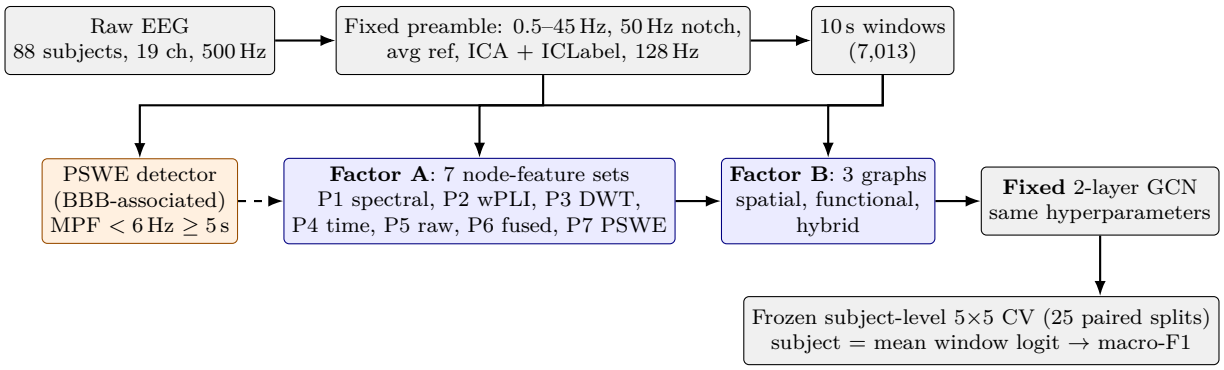


Figure 1: Study design. Grey components are held fixed; only the node features (Factor A) and the graph construction (Factor B) vary, in a fully crossed 7×3 design evaluated on the same 25 frozen subject-level splits. The PSWE detector (orange) supplies the blood–brain-barrier-associated marker, used as P7 node features and, in an ablation, as a co-occurrence graph.

Table 2: Participant characteristics (mean ± SD), from `participants.tsv`. MMSE is reported for description only and is never a model input.

	AD ($n = 36$)	FTD ($n = 23$)	CN ($n = 29$)
Age (years)	66.4 ± 7.9	63.7 ± 8.2	67.9 ± 5.4
Sex (F/M)	24/12	9/14	11/18
MMSE	17.8 ± 4.5	22.2 ± 2.6	30.0 ± 0.0

3.1 Dataset

We used OpenNeuro ds004504 [13, 14]: resting-state, eyes-closed scalp EEG from 88 participants recorded at AHEPA University Hospital (36 AD, 23 FTD, 29 cognitively normal controls, CN). All values below were read from the dataset's own BIDS files rather than from the descriptor paper. Recordings use 19 electrodes of the 10–20 system (Fp1, Fp2, F3, F4, C3, C4, P3, P4, O1, O2, F7, F8, T3, T4, T5, T6, Fz, Cz, Pz) referenced to A1–A2, sampled at 500 Hz on a Nihon Kohden EEG 2100 system, with a 0.4–50 Hz acquisition filter. Recording length per subject ranges from 307 to 1291 s (mean 802 s; 19.6 h in total). Group demographics are given in Table 2. The data are released under CC0; the study uses only de-identified public data. The dataset's own metadata cite a version-1.0.9 DOI that did not resolve when checked; we cite version 1.0.8, whose EEG data are identical (the 1.0.9 changelog records only a README edit).

3.2 Fixed preprocessing

Every condition shares one code path, implemented with MNE-Python [43]. Raw recordings were band-pass filtered at 0.5–45 Hz, notch filtered at 50 Hz, and re-referenced to the common average. Independent component analysis (extended Infomax [44], 18 components — one fewer than the channel count because of the average reference) was fitted on a 1 Hz high-passed copy of each recording, and components that ICLabel [45] assigned to eye or muscle artefact with probability ≥ 0.8 were removed. Data were then resampled to 128 Hz and cut into non-overlapping 10 s windows (1,280 samples), giving 7,013 windows in total (30–129 per subject). ICA is fitted per recording, without labels or other subjects' data, and therefore cannot leak information across cross-validation folds; every transform that is estimated across subjects (feature scaling, demographic residualisation)

is fitted on training subjects only.

3.3 Factor A: node-feature representations

Each window is a graph whose 19 nodes are the electrodes. Seven representations define the node features $X \in \mathbb{R}^{19\times F}$:

P1, spectral ($F = 7$). Relative power in five bands (delta 0.5–4, theta 4–8, alpha 8–13, beta 13–30, gamma 30–45 Hz) from Welch's method [46] (2 s Hann segments, 50% overlap), normalised spectral entropy, and peak frequency within 4–15 Hz.

P2, connectivity ($F = 5$). Node strength of the weighted phase-lag index (wPLI) [37, 38] in each band: the mean wPLI between a channel and the other 18.

P3, wavelet ($F = 20$). Five-level Daubechies-4 discrete wavelet transform [47]; for the sub-bands A5, D5, D4, D3 and D2 (D1, 32–64 Hz, lies above the low-pass cut-off), relative energy, normalised coefficient entropy, mean absolute coefficient and standard deviation.

P4, time domain ($F = 10$). Mean, variance, root mean square, skewness, kurtosis, zero-crossing rate, Hjorth mobility and complexity [48], sample entropy [49] ($m = 2$, $r = 0.2\,\sigma$) and line length. Line length replaces Hjorth activity, which equals the variance.

P5, raw ($F = 1280$). The window z-scored per channel.

P6, fused ($F = 34$). P1, P3 and P2 concatenated, plus each node's weighted clustering coefficient [50] and eigenvector centrality in the alpha-band wPLI graph.

P7, PSWE ($F = 4$). From the PSWE detector (Section 3.8): fraction of the window spent in a PSWE, mean and minimum median power frequency in the window, and the local PSWE onset rate within ±60 s.

The wPLI between channels i and j was computed within each window from band-passed (4th-order Butterworth, zero-phase) analytic signals $z_i(t)$ as

$$\mathrm{wPLI}_{ij} = \frac{\left|\langle \mathrm{Im}(z_i z_j^*)\rangle_t\right|}{\langle\left|\mathrm{Im}(z_i z_j^*)\right|\rangle_t}, \tag{1}$$

which is insensitive to zero-lag coupling from volume conduction [38].

3.4 Factor B: graph construction

Three edge definitions were crossed with every representation. *Spatial*: a Gaussian kernel $w_{ij} = \exp(-d_{ij}^2/2\sigma^2)$ on Euclidean distances between standard 10–20 electrode positions, with σ the median pairwise distance; this graph is identical for every window and subject. *Functional*: the alpha-band wPLI of the window. *Hybrid*: the element-wise product of the two. Each graph was sparsified by keeping the $k = 4$ strongest edges per node, symmetrised as $A = \max(A, A^\top)$, kept weighted, and normalised as $\hat{A} = \tilde{D}^{-1/2}(A + I)\tilde{D}^{-1/2}$.

3.5 The fixed model

The model is the control variable, not the contribution, so a single small graph convolutional network (GCN) [30] was used everywhere: $H_1 = \mathrm{Dropout}_{0.5}(\mathrm{BN}(\mathrm{ReLU}(\hat{A}XW_0)))$, $H_2 = \mathrm{ReLU}(\hat{A}H_1W_1)$, mean pooling over the 19 nodes, and a linear layer to three classes, with $W_0 \in \mathbb{R}^{F\times 64}$ and $W_1 \in \mathbb{R}^{64\times 32}$. This gives 2,819 parameters for $F = 7$ and 3,651 for $F = 20$; P5 is the unavoidable exception (84,355 parameters, because $F = 1280$). Two hops of propagation already reach most of a sparsified 19-node graph, and deeper GCNs risk over-smoothing; depth is tested as an ablation. Graph attention networks [35] learn edge weights and would partly undo the edge definitions being compared, so a GAT is used only as a sensitivity check.

Training used AdamW (learning rate 10^{-3}, weight decay 10^{-4}), batch size 64, and at most 200 epochs with early stopping (patience 20) on validation macro-F1 computed at subject level. The loss was weighted by inverse class frequency (over subjects) divided by each subject's window count, so every subject contributes equally regardless of recording length. A subject's prediction is the mean of its window logits. Features were standardised with statistics from training windows only. All hyperparameters were fixed before any result was seen and are identical in every condition.

3.6 Evaluation protocol

We used subject-level, stratified, repeated 5-fold cross-validation with five repeats (seeds 0–4), for 25 test splits of 17–18 subjects each. Within each training set, a stratified 20% of subjects formed the early-stopping validation set; it was never used to tune hyperparameters. The fold assignment was generated once, verified by assertions that training, validation and test subjects are pairwise disjoint and that every subject is tested exactly once per repeat, and committed to the code repository before any training run. All conditions use the same 25 splits, so every comparison is paired. The full factorial comprises 7×3 cells $\times$ 25 splits = 525 training runs.

Confirmation folds. Choosing the best cell, the cells to ablate and the cell to permute on the same splits that then estimate their performance biases those estimates upwards (the winner's curse). We therefore generated a second, independent fold assignment with the same procedure (seeds 5–9; again 25 splits) and committed it to the repository after the main grid had been analysed but before any run used it. The whole grid was re-run on these folds (525 runs). The cell ranked first on seeds 0–4 is reported with its score on seeds 5–9 as a selection-free estimate, and all ablations, the leakage experiment and the permutation test reported in the main text use the confirmation folds. Their selection-fold counterparts are given in the Supplementary material.

The primary metric is subject-level macro-averaged F1. We also report balanced accuracy, one-vs-rest macro ROC-AUC, PR-AUC, per-class sensitivity and specificity, Cohen's κ, the Brier score, expected calibration error and confusion matrices. Plain accuracy is reported but never used for ranking.

3.7 Baselines, ablations and controls

Baselines. A logistic regression on age and sex alone (the demographic confound floor); a stratified random classifier (chance); and logistic regression, RBF-kernel SVM and random forest trained on the flattened node features of each representation with the same folds, weighting and subject aggregation as the GCN (a non-graph twin of every representation). The MMSE is never used as an input: control participants all score 30, so it is a proxy for the label.

Deep-learning baselines. To place the GCN against standard end-to-end EEG networks, EEGNet-8,2 [51] (temporal convolution of 64 taps, depthwise spatial convolution, separable convolution; 3331 parameters) and a compact 1D-CNN (three temporal convolution blocks over the 19 channels, global average pooling; 47 939 parameters) were trained on the same raw 10 s windows as P5, with the same 25 selection splits, optimiser, early-stopping rule, subject-equal weighting and subject-level aggregation as the GCN, so that every comparison is paired. EEGNet's max-norm weight constraints were replaced by the shared weight decay; no hyperparameter was tuned.

The GCN learns its weights from the training subjects only and uses the validation subjects only to choose when to stop, whereas the classical models have no stopping rule. To compare models trained on the same subjects, we evaluate both under two regimes on the same 25 test splits: (i) *train only*, with the classical models fitted on the training subjects and the GCN as in the main grid; and (ii) *train + val*, with the classical models fitted on training and validation subjects and the GCN refitted on the same subjects for the number of epochs selected by early stopping in the corresponding main-grid run (525 refits, no validation monitoring).

Ablations. Run on the two best cells of the main grid, chosen automatically from the seed 0–4 results and evaluated on the 25 confirmation splits, each paired with the same cell's confirmation-grid run: no graph (identity adjacency, i.e. a per-node MLP); degree-preserving random graphs; $k \in \{3, 6\}$ and a fully connected graph; binarised edges; depth 1 and 3; GAT instead of GCN; no subject-equal weighting; demographic residualisation (age and sex regressed out of every feature, coefficients fitted on training windows); adding P7 (PSWE) features to the best representation; the PSWE co-occurrence graph (Jaccard overlap of channels' PSWE time) as the edge type; 5 s and 20 s instead of 10 s windows (A7; the preprocessed recordings re-windowed and every feature and graph rebuilt); and preprocessing without ICA/ICLabel artefact removal (A8; every other step unchanged). Holm correction is applied within each cell's family of 16 ablations.

Leakage. The best cells were re-run with *window-level* stratified splitting — windows assigned to folds at random, ignoring subject identity (record-wise cross-validation in the terminology of [18]; segment-based holdout in [19]) — with everything else unchanged.

Permutation test. For the best cell, subject labels were permuted (all windows of a subject keep one label) and the first repeat's five folds re-run for each permutation; the p-value is $(1 + \#\{\text{null} \geq \text{observed}\})/(1 + N)$. The primary test uses 100 permutations on the first confirmation repeat (seed 5); the original 200 permutations on the first selection repeat (seed 0) are reported for completeness.

Stopping-rule sensitivity. Early stopping on about 14 validation subjects often selected an early epoch. To test whether this shaped the conclusions, the full 7×3 grid was re-run on the 25 selection splits with patience 50 instead of 20, and with no early stopping at all (100 epochs on the training subjects, final weights kept; validation subjects unused), each paired with the main grid split by split.

3.8 PSWE detection

Following Milikovsky et al. [26], with parameters as reported in [28], we computed the median power frequency (MPF) of each channel from FFTs of 2 s Hann-windowed segments with 1 s overlap (one MPF value per second) over 1–45 Hz. The frequency range is not reported in the source and was set to our pass-band. A PSWE is a run of at least five consecutive seconds with MPF < 6 Hz. Detection

is per channel and per recording, and uses no labels. For each subject we computed the PSWE rate (events per minute, averaged over channels) and the fraction of time spent in PSWEs.

Background-relative PSWE (exploratory). After observing that the fixed-threshold rate was strongly correlated with background slowing (Section 4.9), we added a post-hoc analysis, labelled as such throughout. Its definitions were fixed in the configuration before it was run on any real data. A relative PSWE is a run of at least 5 s in which a channel's MPF lies below that channel's own recording-median MPF by at least 2 Hz (primary), 1.5 Hz or 3 Hz, or by at least 2 or 3 robust standard deviations (1.4826× median absolute deviation).

Detector robustness. Two properties of the fixed-threshold detector could bias it: with 2 s windows the MPF is quantised to 0.5 Hz steps, so the 6 Hz threshold falls on a bin edge; and artefacts add low-frequency power. In a pre-specified robustness analysis we re-detected events with each segment zero-padded to an 8 s FFT (MPF on a 0.125 Hz grid, window unchanged), with amplitude-based artefact rejection (a channel's 2 s segment rejected if its peak-to-peak amplitude exceeded 200 µV or it was flat, SD < 0.5 µV; rejected seconds break events and are removed from the exposure time), and with both (primary), and repeated the statistics below. The unmodified variant reproduced the main event counts exactly in both cohorts. The fixed-threshold statistics were also repeated on the recordings preprocessed without ICA (A8).

3.9 External validation cohort

To test whether the conclusions depend on one cohort, the identical protocol was applied to OpenNeuro ds004584 [52]: 149 participants (100 with Parkinson's disease, 49 controls), 2 min to 5 min of eyes-open resting EEG, 63 channels at 500 Hz. Only what must differ between datasets was changed, in a configuration overlay: the 19 electrodes of the main montage were selected in the same node order, the online reference (Pz), which is not stored, was restored as a flat channel before average re-referencing (exact, because every stored channel is referenced to it); the mains notch was 60 Hz; the task is two-class; and the Montreal Cognitive Assessment replaced the MMSE as the analysis-only cognitive score (the UPDRS, a Parkinson's-only label proxy, is never used). Features, graphs, model, training, cross-validation (seeds 0–4, frozen before any training on this cohort) and statistics were unchanged. Questions were fixed before the first run: whether the ranking of the 21 cells and of the 7 representations transfers (Spearman and Kendall correlations of cell means); how the configuration selected on ds004504 performs without re-selection (with 100 label permutations); whether matched-training non-graph twins, the age-and-sex floor and window-level leakage behave as on ds004504; and whether fixed-threshold PSWEs are more frequent in Parkinson's disease and track slowing, as a test of the earlier report that they are uncommon in this disease [28]. The two datasets differ in site, device, disease, eyes-open versus eyes-closed recording and recording length, so absolute scores are not compared.

3.10 Statistical analysis

Primary inference: by-subject mixed model. A diagnostic claim is about new subjects, so the primary test of the factorial treats the held-out subject as the unit of inference. For every run we take the probability the model assigns to each test subject's true class and average it over the five repeats (each from a different partition), giving one value per subject and cell (1,848 values from 9,240 subject-level predictions for ds004504). These were modelled with a linear mixed model with representation, graph construction and their interaction as fixed effects (sum-to-zero coding), true

class as a covariate, and random intercepts for subject, subject × representation and subject × graph, fitted by REML in statsmodels. The subject-by-factor terms matter: a subject's advantage under one representation recurs in every repeat and graph type, so omitting them treats thousands of correlated predictions as independent (a first version without them gave implausibly small p-values and was discarded). Each term is tested with a Wald F test against its own subject-level stratum (containment degrees of freedom); for balanced data the model reduces to a by-subject repeated-measures ANOVA, which our implementation reproduces exactly in tests, and its false-positive rate is nominal in simulations with subject-by-representation variation. We report model-based marginal means with 95% confidence intervals and Holm-corrected pairwise contrasts between representations. The same model was fitted to the confirmation folds and to the external cohort.

Secondary analyses on macro-F1. The 25 test splits are the repeated-measures units. Main effects of representation and graph construction, and their interaction, were tested with a balanced two-way repeated-measures ANOVA on macro-F1, reporting partial η^2 and Greenhouse–Geisser-corrected p-values. Because cross-validation scores share training data and are therefore not independent, every pairwise comparison is reported with both the Wilcoxon signed-rank test and the Nadeau–Bengio corrected resampled t-test [53], with Holm correction [54] within each family, Cohen's d and the matched-pairs rank-biserial correlation. All 21 cells were also compared with a Friedman test and a Nemenyi critical-difference analysis [55]. Cell-level 95% confidence intervals come from a subject-level bootstrap (2,000 resamples of subjects within each repeat) and from a t-interval over the 25 splits. The Nadeau–Bengio correction uses the ratio of test to training subjects of the models being compared: 17.6/56.3 for models fitted on the training subjects only (the main grid and the ablations) and 17.6/70.4 when validation subjects are added (the train + val regime).

For PSWEs, group differences were tested with Kruskal–Wallis and Dunn post-hoc tests (Holm-corrected, with rank-biserial effect sizes), and with negative-binomial regression of event counts using recording length as exposure, adjusted for age and sex and, in further models, for relative delta+theta power and background MPF. The relation to slowing was assessed with Spearman and partial (age- and sex-adjusted) rank correlations, and the relation to cognition with Spearman correlation against MMSE within patients only. All analyses were run with SciPy, statsmodels and custom code that is tested against reference implementations.

3.11 Compute

All experiments ran on free Kaggle sessions (4 vCPUs); the only GPU use was the two deep-learning baselines (0.34 GPU-hours of training on one Kaggle GPU). The 525-run main grid took 2.4 CPU-hours, and the 2,400 further runs (confirmation grid, ablations and permutations on the confirmation folds, and the train + val refits) took 9.0 CPU-hours. The stopping-rule sensitivity grid and the external cohort added ExtraRuns runs and 13.3 CPU-hours. The mixed models fit in seconds. Every run is logged with its configuration hash, code version and seed.

4 Results

4.1 Data quality and the confound floor

The fixed preprocessing produced 7,013 windows from all 88 subjects (Figure 2); ICA removed on average 2.4 components per recording. The expected EEG slowing in dementia was present before any model was trained: the median theta/alpha power ratio was 1.57 in AD, 0.68 in FTD and 0.28 in controls, with the posterior alpha of controls replaced by diffuse delta and theta in AD (Figure 3).

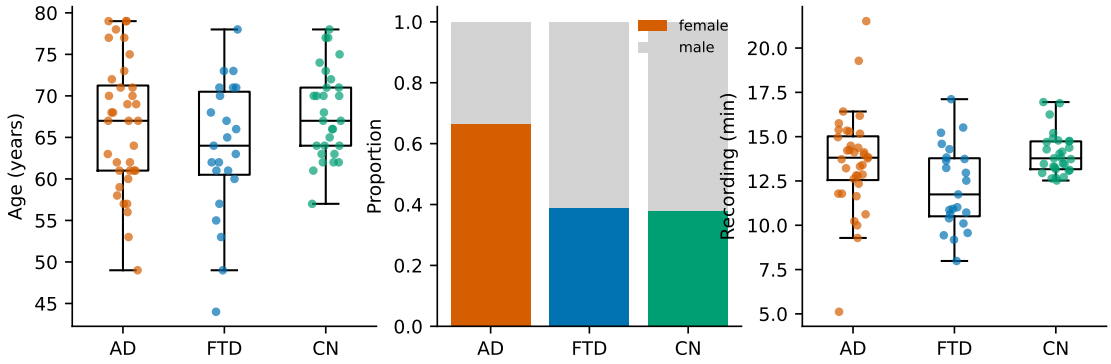


Figure 2: Cohort: age, sex and recording length by group.

Table 3: By-subject linear mixed model (primary analysis) of the probability assigned to each subject's true class: Wald F (numerator, denominator df) and p for each fixed-effect term, on the selection and confirmation folds of ds004504 and on the external cohort ds004584.

	Selection		Confirmation		ds004584	
Term	F (df)	p	F (df)	p	F (df)	p
Representation	16.60 (6, 510)	p <0.001	19.81 (6, 510)	p <0.001	20.03 (6, 882)	p <0.001
Graph construction	8.24 (2, 170)	p <0.001	6.37 (2, 170)	p =0.002	6.21 (2, 294)	p =0.002
Interaction	3.81 (12, 1020)	p <0.001	4.13 (12, 1020)	p <0.001	2.94 (12, 1764)	p <0.001

A logistic regression on age and sex alone reached a macro-F1 of 0.411, against 0.302 for a stratified random classifier. Demographics therefore carry some group information, consistent with the sex imbalance between groups (Table 2). We use 0.411, not 1/3, as the floor that EEG-based models must clear.

4.2 Representation matters; graph construction barely does

Across the 21 cells on the selection folds (525 runs; Table 4), the best configuration was P3 × spatial, with subject-level macro-F1 0.586 ± 0.112 (subject-bootstrap 95% CI 0.538–0.628). Because this cell was chosen as the maximum of 21, that figure is optimistically biased. Re-evaluated on the independent confirmation folds, the same cell scored 0.596 ± 0.089 (95% CI 0.554–0.643; balanced accuracy 0.610, macro ROC-AUC 0.741, Cohen's κ 0.428) and again ranked first of 21. The ordering of all cells was largely preserved (Spearman $\rho = 0.86$ between the two fold sets; Figure S2). We therefore take 0.596 as the selection-free estimate for the best configuration.

In the primary, by-subject mixed model (Table 3, Figure 4), representation had a strong effect on both fold sets (selection $F(6, 510) = 16.60$, $p < 0.001$; confirmation $F = 19.81$, $p < 0.001$). The marginal mean probability assigned to a subject's true class ranged from 0.373 (P2) and 0.391 (P4) to 0.525 (P6) and 0.529 (P3), and 13 of the 21 representation pairs differed after Holm correction on each fold set. Graph construction and its interaction with representation were also detectable at the subject level (selection: graph $F(2, 170) = 8.24$, $p < 0.001$; interaction $F(12, 1020) = 3.81$, $p < 0.001$; confirmation: $p = 0.002$ and $p < 0.001$) but negligible in size: the three graph types' marginal means differed by less than 0.01 (0.461 spatial, 0.453 functional, 0.453 hybrid). Subjects accounted for 55% of the variance and subject × representation for 42%, while subject × graph variance was estimated at zero: individuals differ in which representation suits them, not in which graph does.

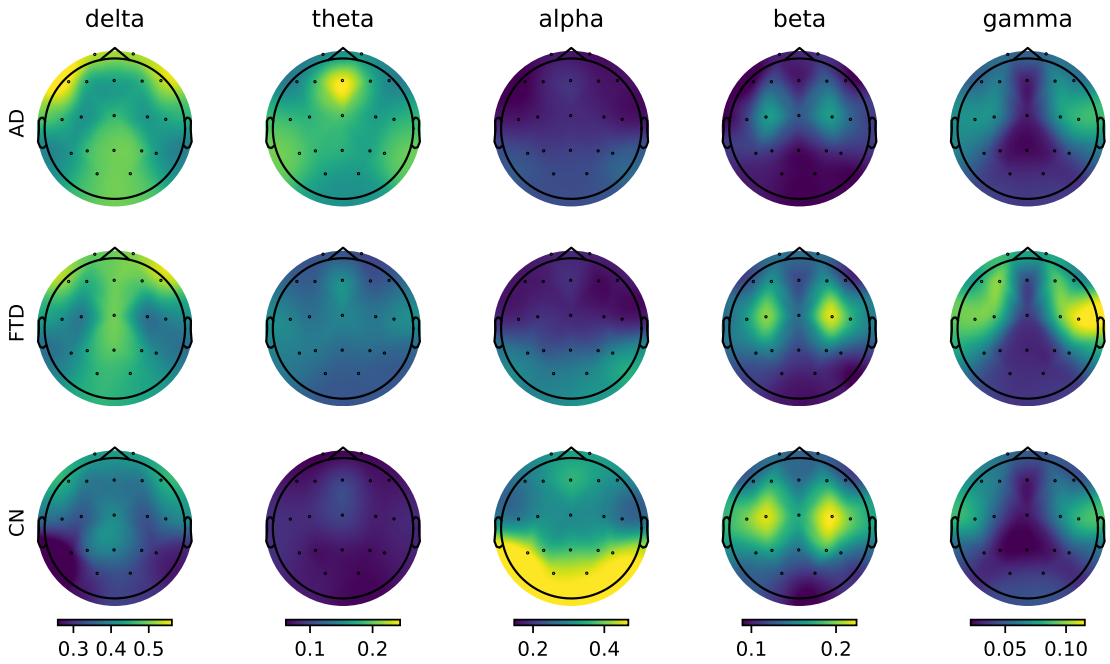


Figure 3: Group-mean relative band power per electrode (rows: groups; columns: bands; colour scales shared within each band).

On macro-F1 with the 25 splits as units (secondary analysis), representation had a large effect on both fold sets (Table 5; selection: $F(6, 144) = 11.83$, Greenhouse–Geisser $p < 0.001$, partial $\eta^2 = 0.33$; confirmation: $F = 13.57$, $p < 0.001$, $\eta^2 = 0.36$). Averaged over graph types, the wavelet (P3, 0.569; confirmation 0.583), spectral (P1, 0.552; 0.528) and fused (P6, 0.543; 0.541) representations formed a leading group; raw windows (P5, 0.489) were intermediate; and connectivity strength (P2, 0.453), PSWE features (P7, 0.448) and time-domain statistics (P4, 0.434) trailed (Figure 5). Graph construction had a small effect on the selection folds ($F(2, 48) = 4.03$, $p = 0.026$, partial $\eta^2 = 0.14$; marginal macro-F1 0.506 spatial, 0.497 functional, 0.492 hybrid; Figure 5) that did not replicate on the confirmation folds ($F = 0.03$, $p = 0.957$; marginal means 0.490, 0.491 and 0.490). The representation × graph interaction was not significant on either (selection $F(12, 288) = 1.43$, $p = 0.194$; confirmation $F = 1.58$, $p = 0.149$; Figure 6). A Friedman test across all cells was significant ($p < 0.001$), but the Nemenyi critical difference (6.26 ranks) left the top cells statistically indistinguishable from one another (Figure 7).

Pairwise comparisons depend on how the dependence between cross-validation splits is handled. With Holm-corrected Wilcoxon tests, 9 of the 21 representation pairs differed on the selection folds and 10 on the confirmation folds; with the more conservative Nadeau–Bengio corrected t-test, 0 and 0 did. For example, P3 exceeded P4 by 0.135 macro-F1 on the selection folds (Cohen's $d = 1.50$; Wilcoxon–Holm $p < 0.001$; Nadeau–Bengio–Holm $p = 0.390$). We regard the omnibus effect and the ordering, which replicated, as robust, and individual pairwise differences as suggestive.

4.3 What the models get right and wrong

For P3 × spatial, sensitivity was 0.771 for controls, 0.642 for AD and only 0.368 for FTD (specificities 0.786, 0.775 and 0.852). Across representations, controls were recognised best and FTD — the smallest group — worst; FTD recordings were most often misclassified as AD, except with PSWE

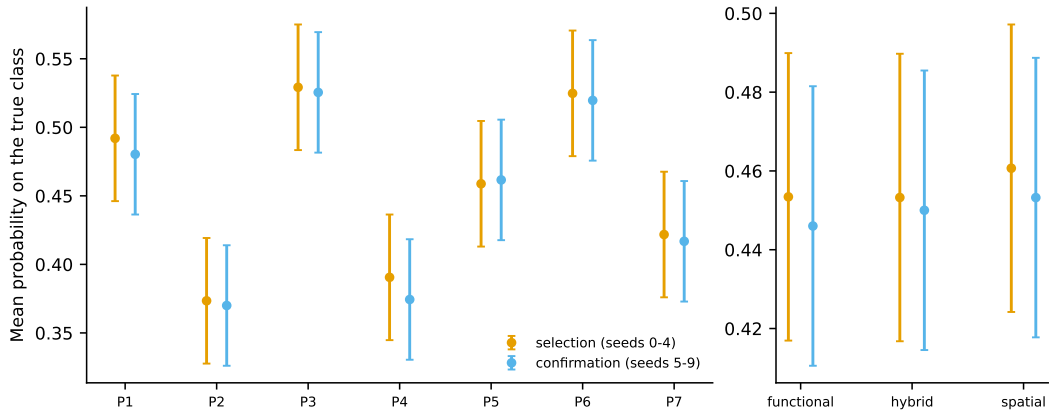


Figure 4: Model-based marginal mean probability on the true class (95% CI) for each representation (left, averaged over graph types) and each graph type (right), from the by-subject mixed model, on the selection and confirmation folds. The intervals include between-subject variation; the tests in Table eftab:mixed compare levels within subjects, which is why a difference much smaller than these intervals can still be detected.

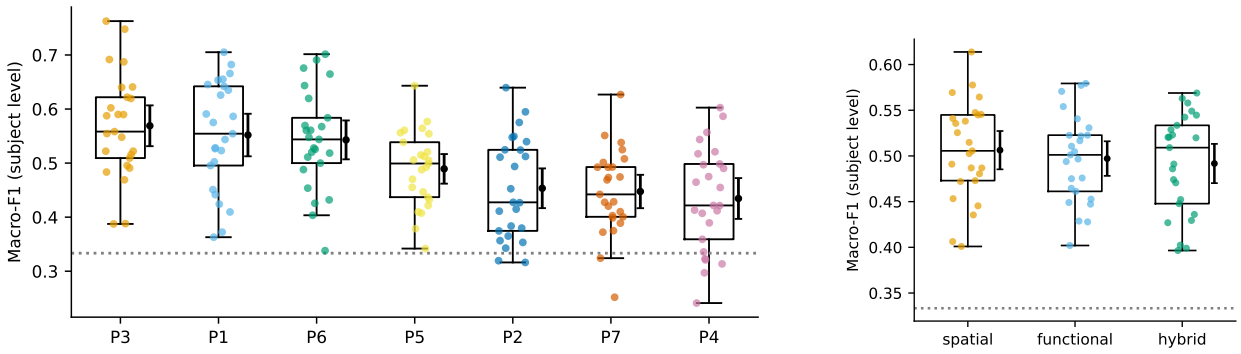


Figure 5: Subject-level macro-F1 by representation (left, averaged over graph types) and by graph construction (right, averaged over representations). Points are the 25 test splits; black markers, mean ± 95% CI; dotted line, 1/3.

features, where they were more often taken for controls (Figure 8). The ROC and precision–recall curves show the same ordering (Figure 9). The models were poorly calibrated (expected calibration error 0.265), so their probabilities should not be read as clinical risks.

4.4 Graph models do not beat their non-graph twins

Table 6 compares every representation's best GCN cell with classical classifiers trained on the same features, test splits and weighting, under both training regimes. With training subjects only, the best classical model reached 0.561; adding the validation subjects raised it to 0.588, and refitting the GCN on the same subjects raised P3 × spatial to 0.613 (mean gain over all cells +0.005). The paired difference between each representation's best GCN cell and its best non-graph twin ranged from −0.044 to +0.050 in the train-only regime and from −0.049 to +0.033 in the train + val regime, and none was significant under either Holm-corrected test (Table S1; for P3, train only: +0.050, Wilcoxon–Holm $p = 0.401$; train + val: +0.033, $p = 0.801$). The GCN on wavelet features was

Table 4: Subject-level performance of every cell (mean ± SD over the 25 selection splits, seeds 0–4). CI: subject-bootstrap 95% interval of macro-F1. BA: balanced accuracy; AUC: one-vs-rest macro ROC-AUC. Confirm.: macro-F1 of the same cell on the 25 independent confirmation splits (seeds 5–9).

Representation	Graph	Macro-F1	95% CI	BA	AUC	κ	Brier	Confirm.
P1 spectral	functional	0.551 ± 0.100	[0.507, 0.603]	0.565	0.724	0.363	0.578	0.537 ± 0.126
P1 spectral	hybrid	0.545 ± 0.117	[0.504, 0.600]	0.561	0.717	0.357	0.599	0.525 ± 0.114
P1 spectral	spatial	0.560 ± 0.103	[0.519, 0.616]	0.575	0.720	0.378	0.593	0.523 ± 0.120
P2 connectivity	functional	0.455 ± 0.108	[0.418, 0.507]	0.477	0.660	0.215	0.626	0.441 ± 0.077
P2 connectivity	hybrid	0.451 ± 0.090	[0.419, 0.507]	0.479	0.659	0.222	0.628	0.445 ± 0.089
P2 connectivity	spatial	0.454 ± 0.102	[0.420, 0.509]	0.483	0.653	0.225	0.629	0.441 ± 0.087
P3 wavelet	functional	0.553 ± 0.106	[0.503, 0.597]	0.560	0.724	0.359	0.598	0.564 ± 0.107
P3 wavelet	hybrid	0.568 ± 0.103	[0.522, 0.614]	0.578	0.735	0.384	0.582	0.588 ± 0.085
P3 wavelet	spatial	0.586 ± 0.112	[0.538, 0.628]	0.594	0.749	0.409	0.558	0.596 ± 0.089
P4 time domain	functional	0.418 ± 0.103	[0.382, 0.473]	0.436	0.623	0.163	0.668	0.435 ± 0.138
P4 time domain	hybrid	0.430 ± 0.104	[0.394, 0.488]	0.450	0.608	0.182	0.677	0.408 ± 0.145
P4 time domain	spatial	0.456 ± 0.099	[0.418, 0.511]	0.474	0.611	0.217	0.677	0.391 ± 0.152
P5 raw	functional	0.504 ± 0.087	[0.469, 0.556]	0.528	0.697	0.320	0.596	0.484 ± 0.120
P5 raw	hybrid	0.470 ± 0.086	[0.435, 0.514]	0.496	0.687	0.275	0.600	0.474 ± 0.089
P5 raw	spatial	0.494 ± 0.097	[0.458, 0.544]	0.526	0.706	0.318	0.594	0.496 ± 0.089
P6 fused	functional	0.556 ± 0.099	[0.506, 0.599]	0.571	0.731	0.370	0.605	0.530 ± 0.101
P6 fused	hybrid	0.532 ± 0.099	[0.490, 0.585]	0.554	0.724	0.342	0.617	0.546 ± 0.112
P6 fused	spatial	0.540 ± 0.094	[0.496, 0.589]	0.558	0.738	0.349	0.589	0.546 ± 0.111
P7 PSWE	functional	0.444 ± 0.085	[0.407, 0.498]	0.467	0.666	0.220	0.615	0.447 ± 0.109
P7 PSWE	hybrid	0.445 ± 0.084	[0.404, 0.495]	0.466	0.670	0.218	0.613	0.442 ± 0.109
P7 PSWE	spatial	0.454 ± 0.082	[0.415, 0.504]	0.472	0.662	0.225	0.620	0.440 ± 0.099

Table 5: Two-way repeated-measures ANOVA on subject-level macro-F1 (units: 25 test splits), on the selection folds and, independently, on the confirmation folds. ε: Greenhouse–Geisser; p: GG-corrected; η_p^2: partial eta squared.

		Selection (seeds 0–4)				Confirmation (seeds 5–9)			
Effect	df	F	ε	p	η_p^2	F	ε	p	η_p^2
Representation	6, 144	11.83	0.71	<0.001	0.33	13.57	0.77	<0.001	0.36
Graph construction	2, 48	4.03	0.96	0.026	0.14	0.03	0.91	0.957	0.00
Interaction	12, 288	1.43	0.60	0.194	0.06	1.58	0.56	0.149	0.06

numerically the best model in both regimes, but its advantage over non-graph classifiers is within the noise of this cohort. The spectral, wavelet, fused and raw representations cleared the demographic floor of 0.411; the connectivity (P2), time-domain (P4) and PSWE (P7) representations scored within about 0.05 of it.

Standard end-to-end EEG networks did no better. On the same raw windows and splits, EEGNet reached 0.530 ± 0.114 (ROC-AUC 0.754) and the 1D-CNN 0.523 ± 0.113 (Table 6; Figure S6), slightly above the GCN given the same raw input (+0.026 and +0.020; Wilcoxon–Holm $p = 0.517$) and below the GCN on wavelet features (-0.056, $p = 0.255$; -0.063, $p = 0.015$). The latter difference was significant for the 1D-CNN under the Wilcoxon test but under neither comparison with the Nadeau–Bengio correction. Learning features end to end from raw signal therefore did not substitute for a good hand-crafted representation on this cohort, while using 0.34 GPU-hours against none for every other model.

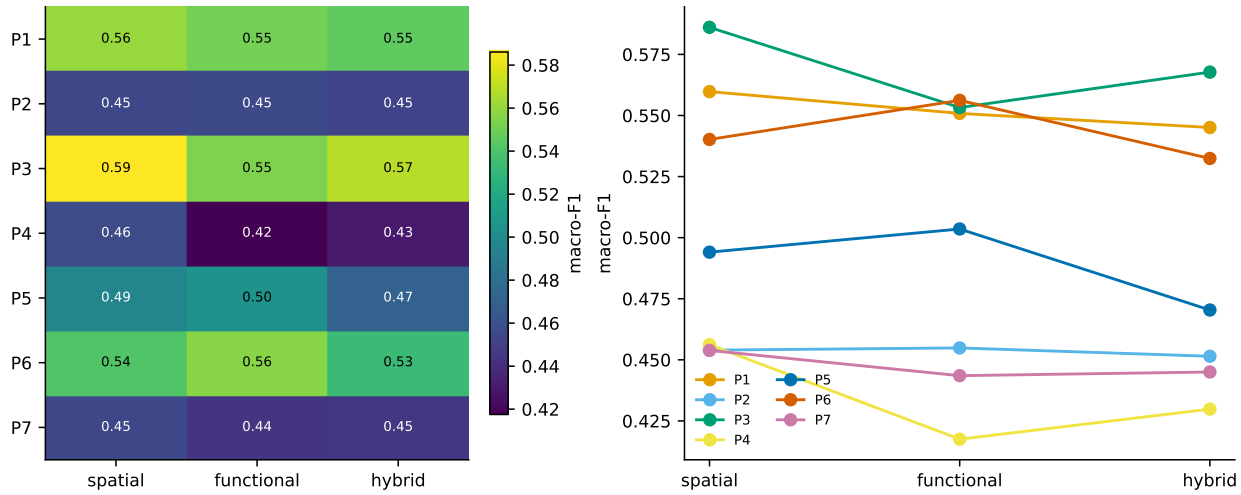


Figure 6: Mean macro-F1 of every representation × graph cell (left) and the same values as an interaction plot (right). Lines are close to parallel: the best representation does not depend on how the graph is built.

4.5 Ablations

Table 8 and Figure 10 report each ablation on the confirmation folds as a paired difference from the same cell's confirmation-grid run. No ablation reached significance under either Holm-corrected test (0 of 32 with Wilcoxon, 0 with Nadeau–Bengio), and every point estimate except one was negative or zero: no modification improved on the reference configuration. Removing the graph altogether (identity adjacency) changed macro-F1 by -0.030 [-0.063, $+0.004$] for the spatial cell and by -0.022 for the hybrid cell, and a degree-preserving random graph by -0.012 and -0.004. Any benefit of message passing is therefore small, and we cannot attribute it to a specific electrode topology rather than to smoothing over neighbours. Two changes had uncorrected t-intervals that excluded zero for the spatial cell: GAT attention (-0.046 [-0.086, -0.006]) and a fully connected graph (-0.032 [-0.065, -0.000]), the latter consistent with over-smoothing; neither interval accounts for the overlap between cross-validation training sets. Sparsity ($k = 3$ or 6), binarised edges, depth and loss weighting changed macro-F1 by at most about 0.02. Regressing age and sex out of every feature cost -0.023 [-0.066, $+0.021$] for the spatial cell, leaving performance well above the demographic floor. Shorter (5 s) or longer (20 s) windows changed macro-F1 by -0.033 and -0.015 for the spatial cell (-0.008 and -0.005 for the hybrid cell), so 10 s windows were never improved upon. Removing ICA artefact rejection cost -0.029 [-0.065, $+0.008$] and -0.035, the largest of the preprocessing changes but still not significant. The selection-fold ablations (Table S2) led to the same conclusions.

4.6 The stopping rule does not drive the results

Early stopping selected a median of epoch 9. Re-running the whole grid with patience 50 (median epoch 17) or with no early stopping (100 epochs, final weights) left the conclusions unchanged (Table 7, Figure S4): P3 × spatial remained the best cell under every rule (0.603 and 0.593), the mean over all cells moved by at most 0.003, cell rankings were preserved (Spearman $\rho = 0.95$ and 0.81 against the main grid), representation stayed significant, and no representation's score changed significantly under either rule (0 of 14 Holm-corrected Wilcoxon tests). Raw windows (P5), the configuration with most parameters, did not benefit from longer training (-0.005 and -0.016), and

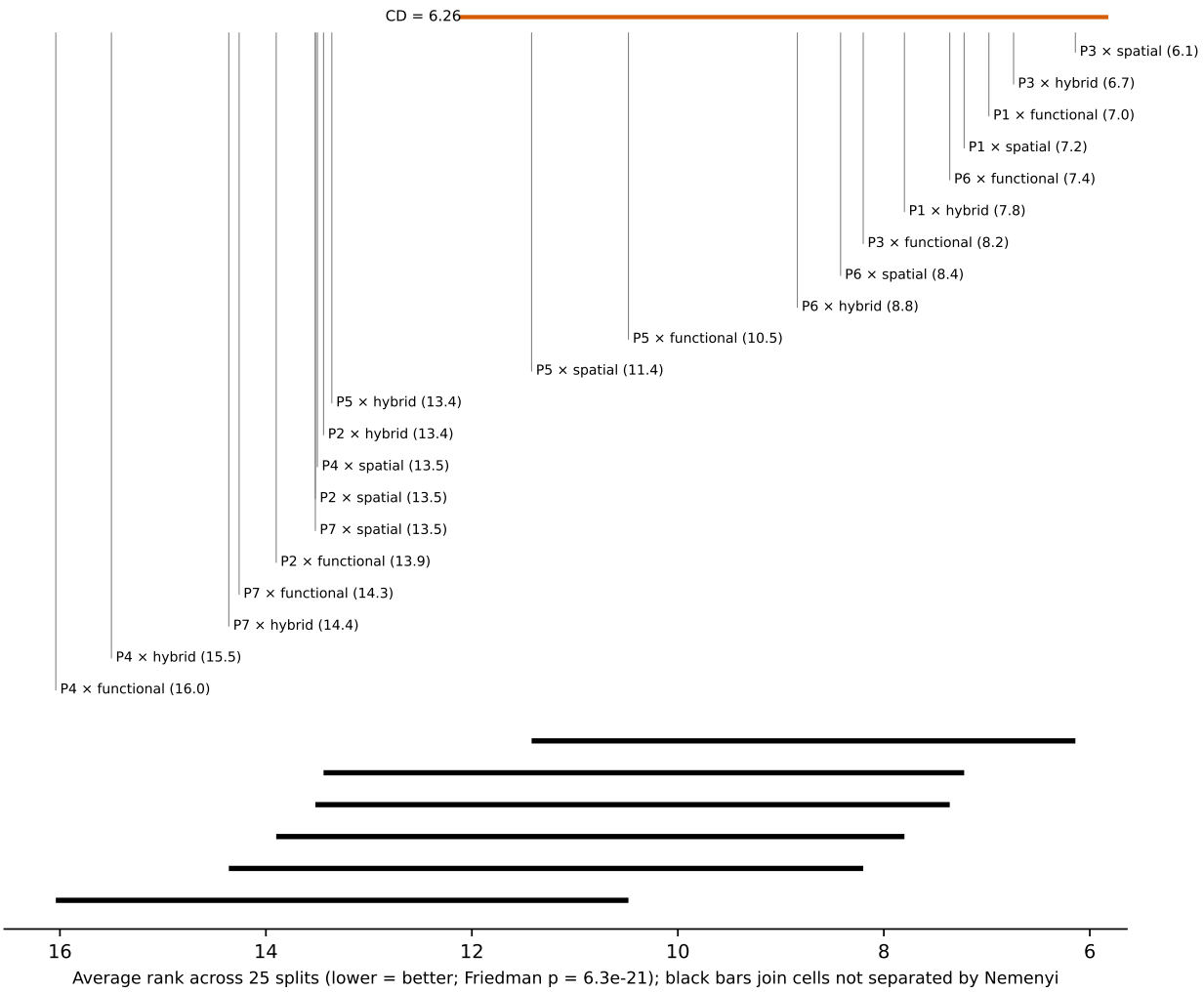


Figure 7: Critical-difference diagram over the 21 cells. Cells joined by a black bar are not separated by the Nemenyi test.

no GCN-versus-classical comparison became significant.

4.7 Leakage inflates results to the published range

Re-running the two best cells on the confirmation folds with window-level splitting — identical data, features, graph, model and hyperparameters — raised subject-level macro-F1 from 0.596 to 0.983 (spatial) and from 0.588 to 0.984 (hybrid). Window-level macro-F1, the metric most often reported, rose to 0.839 and 0.837 (Figure 11). The selection folds gave almost identical values (0.586 to 0.988 for the spatial cell). The leaky protocol thus reproduces the 95–99% figures reported on this benchmark with a model that, evaluated correctly, scores about 0.6.

4.8 The best cell is above chance

On the first confirmation repeat, the observed macro-F1 of P3 × spatial was 0.624, against a label-permutation null with mean 0.312 and maximum 0.461 over 100 permutations ($p = 0.010$, the

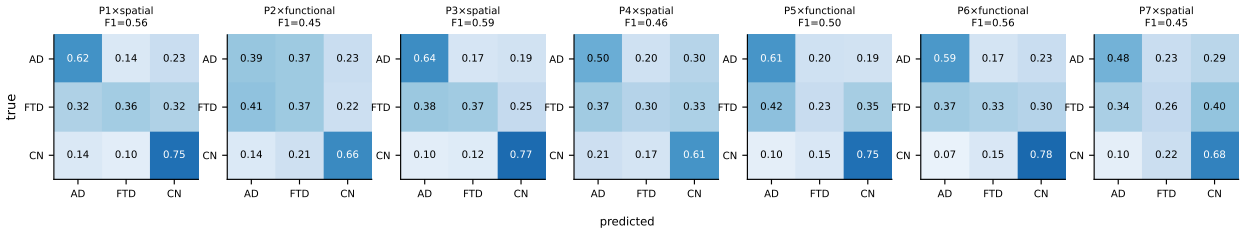


Figure 8: Row-normalised confusion matrices summed over the 25 test splits, for the best graph type of each representation.

smallest value attainable; Figure S3). The original test on the selection folds gave the same result (200 permutations, observed 0.575, null maximum 0.474, $p = 0.005$).

4.9 PSWE: a BBB-associated marker that tracks slowing

Fixed-threshold PSWEs were most frequent in AD (median 1.34 events/min), intermediate in FTD (0.41) and least frequent in controls (0.22; Kruskal–Wallis $p = 0.041$; Figures 12 and 13). In negative-binomial models adjusted for age and sex, the PSWE rate ratio relative to controls was 2.96 (95% CI 1.58–5.57, $p < 0.001$) in AD and 2.07 (1.03–4.15, $p = 0.041$) in FTD.

The PSWE rate was, however, almost collinear with background slowing (Spearman $\rho = 0.88$ with relative delta+theta power; age- and sex-adjusted partial $\rho = 0.87$). Once slowing was added to the model, the AD rate ratio reversed to 0.50 (0.33–0.75, $p < 0.001$), and FTD fell to 0.67 ($p = 0.051$). The PSWE rate was not significantly associated with MMSE within patients ($\rho = -0.22$, $p = 0.095$). In the classifier, PSWE features alone (P7) performed near the demographic floor (0.448), and adding them to the best representation, or using PSWE co-occurrence as the graph, did not help (Section 4.5).

Detector robustness. Neither the coarse MPF grid nor artefacts explain these results. With zero-padded spectra and artefact rejection together (a median of at most 0.3% of segments rejected per group), rates correlated $\rho = 0.98$ with the original ones, the age- and sex-adjusted rate ratios were 4.54 for AD ($p < 0.001$) and 2.85 for FTD ($p = 0.008$), and adjusting for slowing again reversed them (0.50 and 0.62). Without ICA, the rate ratios were 2.35 and 1.99 ($\rho = 0.90$ with slowing), so ocular artefacts do not create the excess either.

Background-relative PSWEs (exploratory). Defining PSWEs relative to each channel's own background reversed the group pattern. With the primary definition (a drop of at least 2 Hz below the channel median), AD showed *fewer* transient slowing events than controls (rate ratio 0.55, 0.35–0.89, $p = 0.014$), FTD a non-significant reduction (0.70), and the event rate was only weakly related to slowing ($\rho = -0.15$) and not to MMSE ($\rho = 0.04$). After additional adjustment for background MPF the AD effect was no longer significant (0.77, $p = 0.299$). The MAD-scaled variants gave the same direction more strongly and survived that adjustment (for 2 MAD: AD 0.43, $p = 0.014$; FTD 0.35, $p = 0.002$). Every variant pointed the same way: relative to their own background, patients with dementia had fewer, not more, transient slowing events.

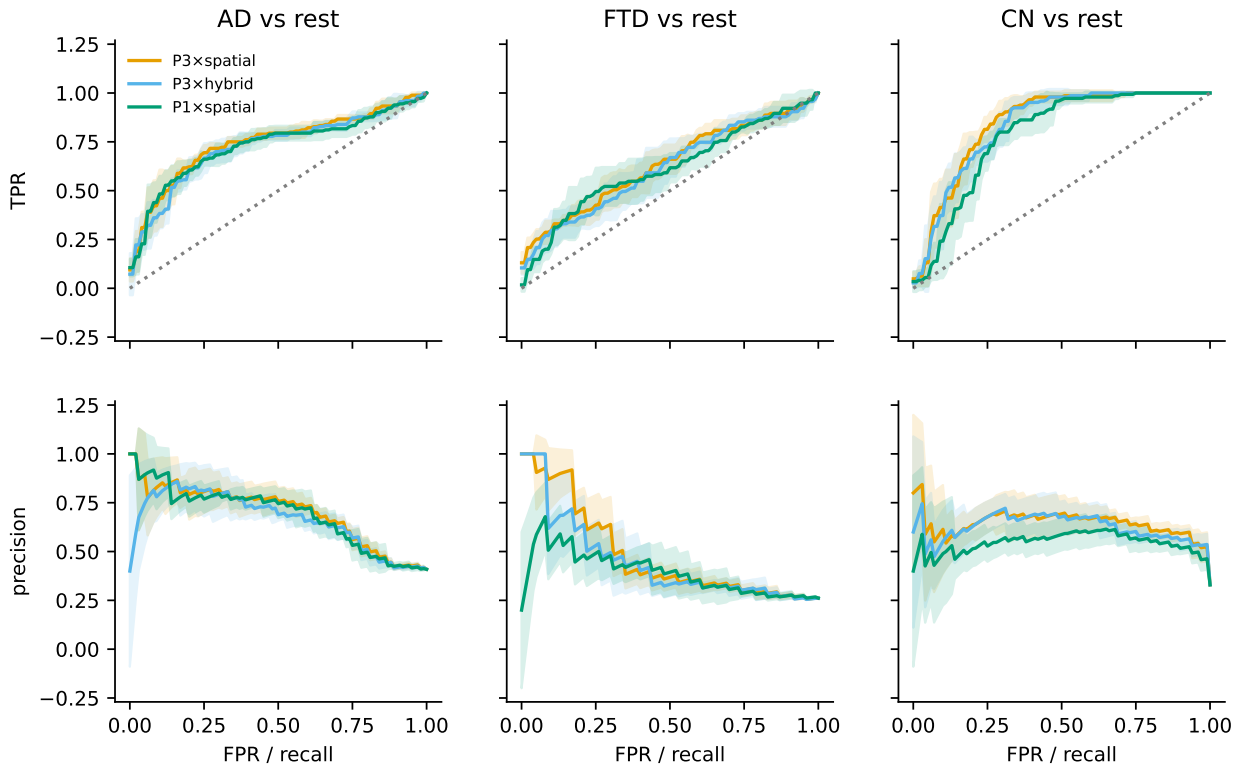


Figure 9: One-vs-rest ROC (top) and precision–recall (bottom) curves for the three best cells; mean ± SD across the five repeats.

4.10 External validation: Parkinson's disease (ds004584)

The ds004584 recordings passed the same quality gates (2,327 windows from 149 subjects; 2.2 ICA components removed per recording on average), and theta/alpha slowing was again higher in patients (0.95) than in controls (0.72). Age and sex alone reached a macro-F1 of 0.550 against 0.490 for chance in this two-class problem.

Protocol effects replicated. Representation again dominated: in the by-subject mixed model, $F(6, 882) = 20.03$, $p < 0.001$, with 13 of 21 pairs separated, whereas graph construction was small (marginal means 0.587 spatial, 0.593 functional, 0.590 hybrid; $p = 0.002$) and favoured a different graph type than on ds004504; on macro-F1 with splits as units, representation was significant ($F = 23.19$, $p < 0.001$, partial $\eta^2 = 0.49$) and graph construction was not ($p = 0.207$). The GCN was not significantly better than non-graph classifiers trained on the same subjects (differences -0.008 to $+0.049$, none significant after Holm correction). Window-level splitting of the configuration selected on ds004504 inflated subject-level macro-F1 from 0.688 to 0.839 (window-level 0.810), in line with an independent audit of the same cohort [20].

The ranking only partly transferred. The configuration selected on ds004504 (P3 × spatial), evaluated without re-selection, reached macro-F1 0.688 ± 0.101 (subject-bootstrap 95% CI 0.651–0.721; balanced accuracy 0.709, ROC-AUC 0.766), well above its permutation null (observed 0.679 on the first repeat against a null maximum of 0.593; $p = 0.010$, 100 permutations) and ranked 5th of 21; the best cell was P6 × hybrid (0.708). Wavelet and fused features were among the top three on

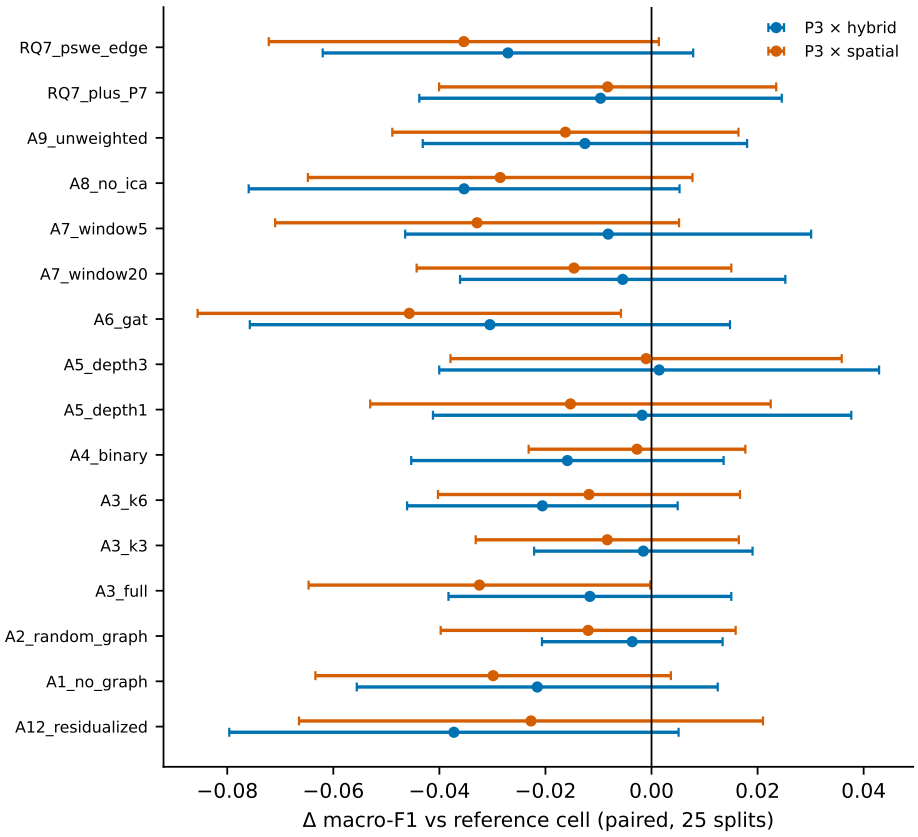


Figure 10: Ablations on the confirmation folds as paired differences from the two reference cells (mean and 95% CI over 25 splits). An asterisk would mark Nadeau–Bengio–Holm $p < 0.05$; none qualifies.

both datasets (Table 9), but the full ordering did not transfer: Spearman $\rho = 0.35$ ($p = 0.122$) across the 21 cells and $\rho = 0.29$ ($p = 0.535$) across the seven representations, mainly because time-domain (P4) and connectivity (P2) features, near the bottom on ds004504, ranked third and fourth on ds004584, and raw windows (P5) fell to last (Figure 14).

PSWEs in Parkinson's disease. Fixed-threshold PSWEs were not more frequent in Parkinson's disease than in controls (median 0.09 versus 0.10 events/min; age- and sex-adjusted rate ratio 1.21, 95% CI 0.65–2.23, $p = 0.552$), consistent with the earlier report that they are uncommon in this disease [28]. As on ds004504, the rate tracked background slowing ($\rho = 0.81$), and adjusting for slowing reversed the ratio (0.57, $p = 0.005$). The rate was not associated with MoCA within patients ($\rho = -0.11$, $p = 0.297$), no background-relative variant differed between groups (smallest $p = 0.186$), and zero-padding with artefact rejection left the result unchanged (rate ratio 1.17, $p = 0.674$).

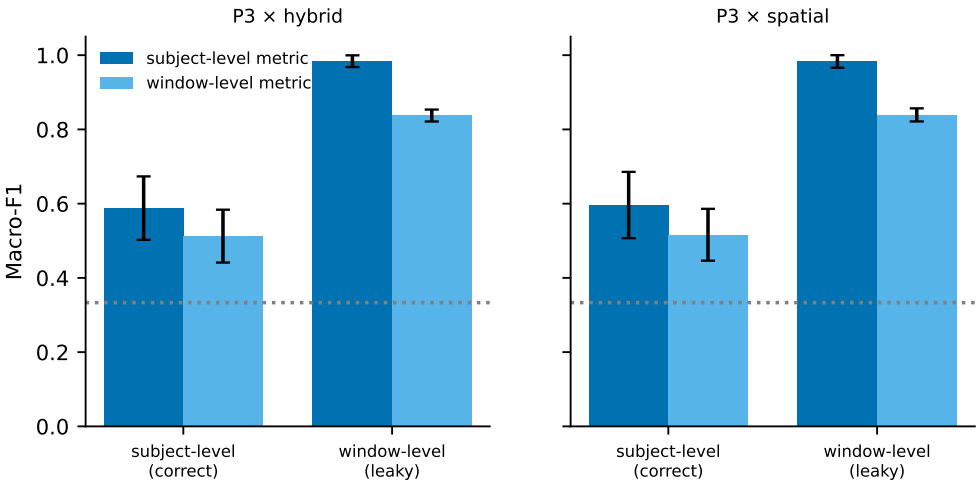


Figure 11: The same pipeline evaluated with subject-level (correct) and window-level (leaky) splitting on the confirmation folds. Bars: mean ± SD over 25 splits; dotted line, 1/3.

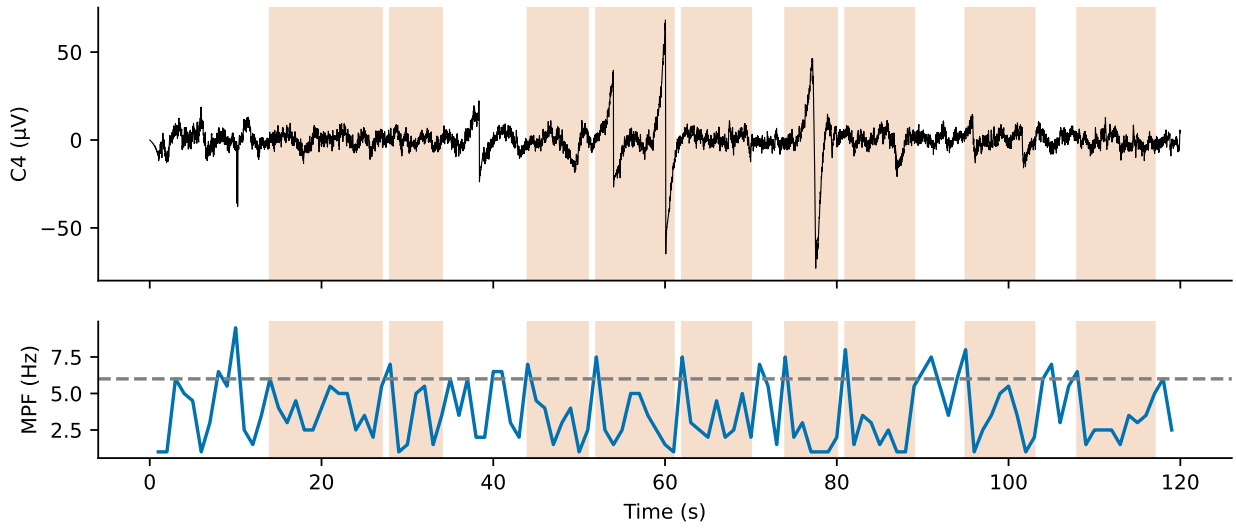


Figure 12: Fixed-threshold PSWE detection in the AD recording with the highest PSWE rate among the example shard (first 120 s, channel with most events). Top: EEG; bottom: median power frequency with the 6 Hz threshold; shaded, detected PSWEs. Because this subject's background is already slow, most of the recording is flagged — the pattern behind the collinearity with slowing.

Table 6: Every representation's best GCN cell against its non-graph twins (same test splits, weighting and subject aggregation) under both training regimes, and the demographic floor. Macro-F1, mean ± SD over the 25 selection splits. Train only: models fitted on the training subjects (the GCN early-stopped on the validation subjects). Train + val: models fitted on training and validation subjects (the GCN refitted for its early-stopped epoch count). GCN graph type: best in each regime.

Representation	Model	Train only	Train + val
P1 spectral	GCN (spatial / functional)	0.560 ± 0.103	0.558 ± 0.110
	Logistic regression	0.536 ± 0.110	0.564 ± 0.096
	RBF-SVM	0.517 ± 0.126	0.560 ± 0.108
	Random forest	0.541 ± 0.116	0.588 ± 0.117
P2 connectivity	GCN (functional / spatial)	0.455 ± 0.108	0.453 ± 0.089
	Logistic regression	0.418 ± 0.083	0.417 ± 0.082
	RBF-SVM	0.459 ± 0.104	0.435 ± 0.089
	Random forest	0.437 ± 0.085	0.418 ± 0.055
P3 wavelet	GCN (spatial / spatial)	0.586 ± 0.112	0.613 ± 0.112
	Logistic regression	0.441 ± 0.132	0.466 ± 0.126
	RBF-SVM	0.513 ± 0.131	0.543 ± 0.122
	Random forest	0.536 ± 0.105	0.581 ± 0.092
P4 time domain	GCN (spatial / spatial)	0.456 ± 0.099	0.439 ± 0.097
	Logistic regression	0.441 ± 0.121	0.446 ± 0.112
	RBF-SVM	0.437 ± 0.109	0.483 ± 0.103
	Random forest	0.481 ± 0.119	0.487 ± 0.123
P5 raw	GCN (functional / spatial)	0.504 ± 0.087	0.484 ± 0.082
	EEGNet	0.530 ± 0.114	–
	1D-CNN	0.523 ± 0.113	–
P6 fused	GCN (functional / spatial)	0.556 ± 0.099	0.580 ± 0.097
	Logistic regression	0.525 ± 0.129	0.514 ± 0.118
	RBF-SVM	0.561 ± 0.092	0.579 ± 0.098
	Random forest	0.540 ± 0.107	0.578 ± 0.118
P7 PSWE	GCN (spatial / spatial)	0.454 ± 0.082	0.456 ± 0.079
	Logistic regression	0.457 ± 0.108	0.456 ± 0.066
	RBF-SVM	0.485 ± 0.102	0.473 ± 0.088
	Random forest	0.498 ± 0.082	0.502 ± 0.077
Age + sex	Logistic regression	–	0.411 ± 0.120
	Stratified chance	–	0.302 ± 0.110

Table 7: Stopping-rule sensitivity on the 25 selection splits: median epoch of the kept weights, best cell, macro-F1 of P3 × spatial, mean over the 21 cells, split-level ANOVA F for representation and graph, and Spearman correlation of the 21 cell means with the main grid.

Stopping rule	Epoch	Best cell	P3 × spatial	Mean	F_{rep}	F_{graph}	ρ
patience 20 (main)	9	P3 × spatial	0.586	0.498	11.83 (p <0.001)	4.03 (p =0.026)	1.00
patience 50	17	P3 × spatial	0.603	0.498	10.66 (p <0.001)	3.78 (p =0.032)	0.95
fixed 100 epochs	100	P3 × spatial	0.593	0.501	7.45 (p <0.001)	7.57 (p =0.002)	0.81

Table 8: Ablations on the confirmation folds: paired Δ macro-F1 (ablated minus the same cell's confirmation-grid run) with 95% t-interval over the 25 splits, and Nadeau–Bengio p with Holm correction within each cell. Selection-fold values: Table S2.

	P3 × spatial		P3 × hybrid	
Ablation	Δ [95% CI]	p_{Holm}	Δ [95% CI]	p_{Holm}
Age/sex regressed out	-0.023 [-0.066, +0.021]	1.00	-0.037 [-0.080, +0.005]	1.00
No graph (identity)	-0.030 [-0.063, +0.004]	1.00	-0.022 [-0.056, +0.012]	1.00
Random graph (degree-preserving)	-0.012 [-0.040, +0.016]	1.00	-0.004 [-0.021, +0.013]	1.00
Fully connected	-0.032 [-0.065, -0.000]	1.00	-0.012 [-0.038, +0.015]	1.00
$k = 3$	-0.008 [-0.033, +0.016]	1.00	-0.002 [-0.022, +0.019]	1.00
$k = 6$	-0.012 [-0.040, +0.017]	1.00	-0.021 [-0.046, +0.005]	1.00
Binarised edges	-0.003 [-0.023, +0.018]	1.00	-0.016 [-0.045, +0.014]	1.00
Depth 1	-0.015 [-0.053, +0.022]	1.00	-0.002 [-0.041, +0.038]	1.00
Depth 3	-0.001 [-0.038, +0.036]	1.00	+0.001 [-0.040, +0.043]	1.00
GAT instead of GCN	-0.046 [-0.086, -0.006]	1.00	-0.030 [-0.076, +0.015]	1.00
20 s windows	-0.015 [-0.044, +0.015]	1.00	-0.005 [-0.036, +0.025]	1.00
5 s windows	-0.033 [-0.071, +0.005]	1.00	-0.008 [-0.046, +0.030]	1.00
No ICA artefact removal	-0.029 [-0.065, +0.008]	1.00	-0.035 [-0.076, +0.005]	1.00
No subject-equal weighting	-0.016 [-0.049, +0.016]	1.00	-0.013 [-0.043, +0.018]	1.00
+ P7 PSWE features	-0.008 [-0.040, +0.024]	1.00	-0.010 [-0.044, +0.025]	1.00
PSWE co-occurrence edges	-0.035 [-0.072, +0.001]	1.00	-0.027 [-0.062, +0.008]	1.00

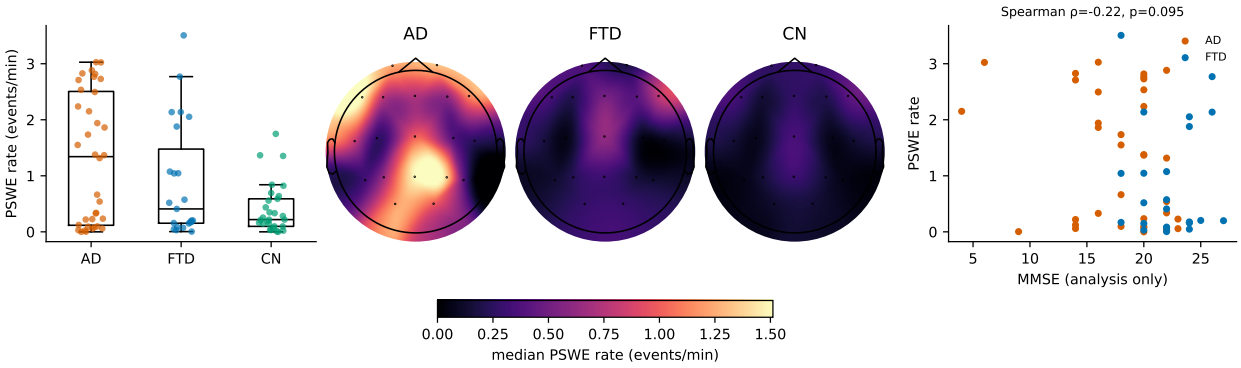


Figure 13: Fixed-threshold PSWE rate per subject by group (left), median per-channel rate as scalp maps (middle), and rate against MMSE in patients (right; analysis only).

Table 9: Mean macro-F1 of each representation (averaged over graph types) and its rank on ds004504 (AD/FTD/CN, three classes) and on the external cohort ds004584 (PD/CN, two classes). Absolute values are not comparable across datasets; ranks are.

	ds004504		ds004584	
Representation	Macro-F1	Rank	Macro-F1	Rank
P1 spectral	0.552	2	0.594	5
P2 connectivity	0.453	5	0.614	4
P3 wavelet	0.569	1	0.689	2
P4 time domain	0.435	7	0.659	3
P5 raw	0.489	4	0.543	7
P6 fused	0.543	3	0.702	1
P7 PSWE	0.447	6	0.555	6

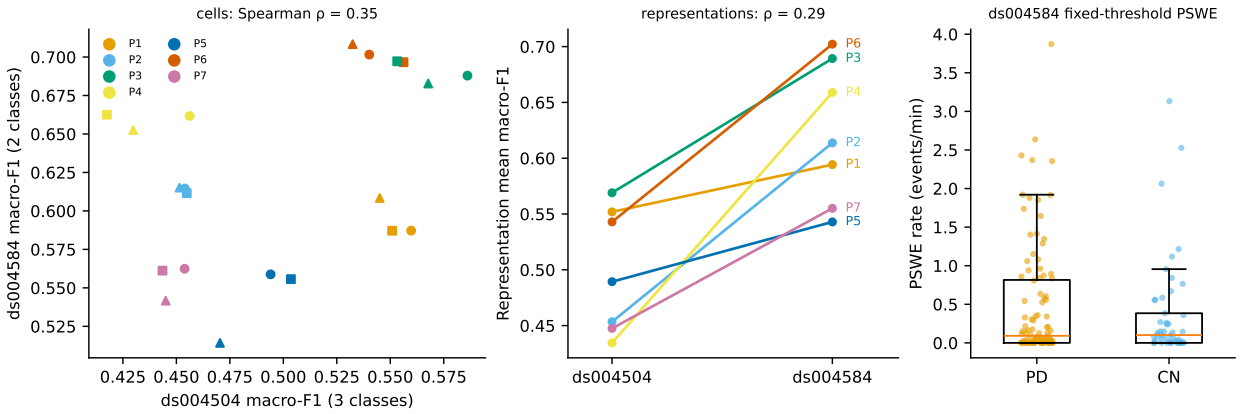


Figure 14: External validation on ds004584. Left: every cell's mean macro-F1 on ds004504 against ds004584. Middle: representation means on both datasets. Right: fixed-threshold PSWE rate per subject in Parkinson's disease and controls.

5 Discussion

5.1 Principal findings

Holding the model, data and evaluation protocol fixed, we found four things. First, the node-feature representation strongly affects three-class AD/FTD/control classification, with subjects as the unit of inference ($F = 16.60$ and 19.81 on two independent fold sets) and on split-level macro-F1 (partial $\eta^2 = 0.33$ and 0.36), with wavelet, spectral and fused features leading. Second, how the graph is built matters little: its effect on macro-F1 did not replicate across fold sets, and although it is detectable per subject, the three graph types differ by less than 0.01 in the probability assigned to the true class, in a direction that differs between cohorts; the GCN is not significantly better than a random forest or SVM trained on the same features and the same subjects, and end-to-end deep EEG networks (EEGNet, a 1D-CNN) on raw windows did not reach it either. These conclusions held under three stopping rules, with 5 s to 20 s windows and without ICA. Third, splitting windows instead of subjects inflates subject-level macro-F1 from about 0.6 to about 0.98 on the identical pipeline, reproducing the top of the published range. Fourth, the blood–brain-barrier-associated PSWE marker is more frequent in AD and FTD than in controls, but this excess is explained by continuous background slowing; defined relative to each person's background, transient slowing events are, if anything, less frequent in dementia.

5.2 Representation matters more than topology

The size and consistency of the representation effect, against a small and non-interacting graph effect, indicate that the information a GNN can use on this benchmark lives mostly in the node features. Several observations point the same way. Spatial graphs — which are identical for every subject and therefore carry no diagnostic information of their own — did at least as well as functional and hybrid graphs, and on the confirmation folds the three graph types were indistinguishable. Neither removing the graph nor replacing it with a degree-preserving random graph changed performance significantly; any benefit of message passing is small, and we cannot attribute it to a particular connectivity pattern rather than to smoothing over neighbouring electrodes. And non-graph classifiers on flattened features, trained on the same subjects, were statistically indistinguishable from the GCN for every representation, although the GCN on wavelet features was numerically best in both training regimes. These results agree with the independent conclusion of the 2026 scoping review that apparent representational gains on this dataset do not survive leakage-free evaluation [15], and they temper the case for graph-based modelling made in recent work [21–23]: on 19-electrode scalp EEG from 88 subjects, we see no evidence that graph structure adds information beyond good node features. Whether this holds with source-space nodes, learned adjacency or much larger cohorts remains open.

5.3 Neurophysiological interpretation

Representations that describe each electrode's own spectrum (P1, P3) outperformed one built only from inter-electrode phase coupling (P2), and the functional graph did not improve on the spatial one. Within the limits of scalp EEG, this favours local oscillatory slowing over altered large-scale coupling as the dominant discriminative signal, consistent with spectral slowing being the most replicated EEG marker of AD [31, 32] and with the posterior-alpha-to-diffuse-theta shift visible in our data (Figure 3). It does not contradict network accounts of dementia [1, 2]: alpha-band wPLI estimated within 10 s windows is noisy, and electrode-level graphs are a coarse proxy for cortical networks. The error pattern is clinically recognisable: controls are identified best, while FTD —

the smallest group, and less slowed than AD — is identified worst and is usually mistaken for AD. Separating the two dementias remains the hard part of the problem.

5.4 What the PSWE results mean for an EEG marker of BBB dysfunction

With the fixed threshold, our data agree in direction with Milikovsky et al.: PSWE rates were roughly three times higher in AD than in controls after adjusting for age and sex, and FTD — not previously studied — was also elevated. Two further analyses change the interpretation. The fixed-threshold rate was almost collinear with relative delta+theta power, and adjusting for that slowing reversed the AD effect. When events were instead defined as transient drops below each channel's own background, AD and FTD showed fewer of them than controls, under every variant we examined. Together these results suggest that, in resting scalp EEG with diffuse slowing, a median power frequency below 6 Hz mostly flags a slow *background* rather than *paroxysmal* events.

We draw three cautious conclusions. First, a fixed-threshold PSWE rate should not be read as evidence of BBB dysfunction without controlling for background slowing. Second, background-relative definitions are needed if PSWEs are to capture transient cortical dysfunction, although they bring their own floor effect in very slow recordings, which we observed in the Hz-drop variants. Third, our data cannot test the BBB link itself: ds004504 contains no BBB imaging, the original detection details include parameters we could not verify from the full text, and Milikovsky et al. studied different patients under different recording conditions. The decisive test is a cohort with both EEG and dynamic contrast-enhanced MRI.

5.5 Implications for how results on this benchmark are read

On one pipeline, the choice between subject-level and window-level splitting moved subject-level macro-F1 by about 0.4 — roughly three times the gap between the best and worst representation, and several times larger than any graph or architecture change we tested. Figures in the high 90s on ds004504 should therefore be read alongside their splitting protocol, and studies that report window-level metrics or allow a subject's windows into both training and test sets cannot be compared with subject-level results. We name no individual study: the point is that the protocol, not the model, determines most of the reported number, as argued independently for this benchmark [15], for translational EEG [19, 20, 40] and for machine-learning-based science generally [16–18]. The permutation test and the independent confirmation folds show that the honest result is nonetheless real and is not an artefact of choosing the best of 21 configurations.

5.6 What the external cohort adds

Applying the identical protocol to Parkinson's disease, a different site, device and recording state, reproduced the protocol-level findings — a dominant representation effect, a negligible graph effect, no advantage of the GCN over non-graph twins, inflation under window-level splitting, and a fixed-threshold PSWE rate that tracks slowing — but not the detailed ranking of representations ($\rho = 0.29$). The configuration chosen on dementia generalised reasonably (0.688, 5th of 21), and wavelet and fused features were near the top in both cohorts, but time-domain and connectivity features, weak for dementia, were competitive for Parkinson's disease. This argues against treating any representation ranking from one disease as a general recommendation: the evaluation protocol transfers; the best representation is disease- and possibly recording-specific.

5.7 Limitations

The cohort is small, and cross-validation estimates in samples of this size carry wide error bars [42]: test folds contain 17–18 subjects, and under the conservative Nadeau–Bengio correction no pairwise difference between representations, and no single ablation, reached significance. We report both corrected and uncorrected tests and give effect sizes, but small differences between leading representations should not be over-interpreted. All data come from one site, device and protocol, and FTD is the smallest class ($n = 23$). The external cohort addresses generalisation of the protocol but differs in disease, task (eyes open), mains frequency and recording length (median about 2.5 min), records no medication state, and is two-class, so it cannot test whether the AD/FTD results themselves replicate. The confirmation folds re-sample the same 88 subjects, so they remove the selection bias of choosing among configurations but are not an external replication. Hyperparameters were fixed a priori and not tuned. Early stopping on a validation set of about 14 subjects selected a median of 9 epochs (23% of runs stopped at epoch 3 or earlier; median 5 for P1 and 20 for P3), so many models are lightly trained; however, training longer (patience 50, or 100 epochs without early stopping) changed no conclusion and did not help the highest-capacity configuration (raw windows, about 84k parameters), and refitting on training plus validation subjects for the selected epoch count changed macro-F1 by only +0.005 on average. Calibration was poor. Graphs are defined on electrodes rather than cortical sources. The demographic floor (0.411) shows residual age and sex information, which residualisation reduced but did not remove. For PSWEs, the MPF frequency range was not reported in the source and was set by us (zero-padding the spectrum and rejecting artefacts did not change the results), we detected events per channel rather than over averaged regions, the background-relative analysis is post hoc, and no BBB measure is available.

5.8 Future work

The most valuable next study is a multi-site replication of this exact protocol, since a single-site design cannot establish device or population invariance. Natural extensions are source-space graphs [23], learned adjacency [8], spatio-temporal graphs over window sequences [9, 11], an architecture sweep under a fixed representation (the transpose of this design) [6], an MCI class, and — for the BBB question — simultaneous EEG and dynamic contrast-enhanced MRI to test whether any PSWE definition tracks measured barrier leakage [24, 25].

6 Conclusion

We asked whether the performance of graph neural networks for EEG-based dementia classification comes from how the signal is represented, how the brain graph is built, or the model. On a public 88-subject benchmark, with the model, data and evaluation protocol held fixed and seven representations crossed with three graph constructions over 25 paired subject-level splits, the answer is mostly the representation: wavelet and spectral node features lead, graph construction matters little, and the GCN does not outperform non-graph classifiers on the same features. These conclusions held with subjects as the unit of inference, on independent splits, under different stopping rules and preprocessing choices, and — at the level of the protocol — in an external Parkinson's disease cohort, although which representation works best did not transfer between diseases. The best configuration, chosen on one set of splits and re-evaluated on an independent set, reached an honest three-class macro-F1 of 0.596, clearly above chance ($p = 0.010$) but far below published figures — a gap that window-level splitting alone closes on our own pipeline, inflating subject-level macro-F1 to 0.983. A blood–brain-barrier-associated EEG marker, the paroxysmal slow-wave event, was elevated in AD

and FTD under its original definition but was explained by background slowing, was not elevated in Parkinson's disease, and reversed when defined relative to each person's background. We release the code, seeds, frozen folds and every raw run so that each number in this paper can be regenerated from the data.

Data and code availability

The EEG data are OpenNeuro ds004504 (CC0) [14]. Code, configuration (`configs/default.yaml`), the frozen fold assignment (`splits/folds_ds004504.json`, with the SHA-256 of `participants.tsv`), every raw training run (one JSON record per run, with configuration hash, code version, seed and per-subject test predictions), and the Kaggle notebooks that produced them are available at `https://github.com/Devguru-codes/Major_project_IIITN`. Random seeds are 0–4 for the main grid (selection folds), 5–9 for the confirmation folds (`splits/folds_ds004504_confirm.json`, committed before any confirmation run), and $10{,}000 + i$ and $11{,}000 + i$ for permutation i on the selection and confirmation folds. Every table and figure in this paper is regenerated from the committed raw runs by a single command (`python -m eegrep report`), and the resolved package versions are recorded in `reports/kaggle_pip_freeze.txt`. The full study runs on free cloud CPUs.

References

[1] C. Stam, B. Jones, G Nolte, M Breakspear, and P. Scheltens. Small-world networks and functional connectivity in alzheimer's disease. *Cerebral Cortex*, 17(1):92–99, 2006. ISSN 1460-2199. doi: 10.1093/cercor/bhj127. URL `http://dx.doi.org/10.1093/cercor/bhj127`.

[2] Willem de Haan, Yolande AL Pijnenburg, Rob LM Strijers, Yolande van der Made, Wiesje M van der Flier, Philip Scheltens, and Cornelis J Stam. Functional neural network analysis in frontotemporal dementia and alzheimer's disease using eeg and graph theory. *BMC Neuroscience*, 10(1), 2009. ISSN 1471-2202. doi: 10.1186/1471-2202-10-101. URL `http://dx.doi.org/10.1186/1471-2202-10-101`.

[3] Cosimo Ieracitano, Nadia Mammone, Amir Hussain, and Francesco C. Morabito. A novel multi-modal machine learning based approach for automatic classification of eeg recordings in dementia. *Neural Networks*, 123:176–190, 2020. ISSN 0893-6080. doi: 10.1016/j.neunet.2019.12.006. URL `http://dx.doi.org/10.1016/j.neunet.2019.12.006`.

[4] Andreas Miltiadous, Emmanouil Gionanidis, Katerina D. Tzimourta, Nikolaos Giannakeas, and Alexandros T. Tzallas. Dice-net: A novel convolution-transformer architecture for alzheimer detection in eeg signals. *IEEE Access*, 11:71840–71858, 2023. ISSN 2169-3536. doi: 10.1109/access.2023.3294618. URL `http://dx.doi.org/10.1109/access.2023.3294618`.

[5] Rundong Jiang, Xiaowei Zheng, Jiamin Sun, Lei Chen, Guanghua Xu, and Rui Zhang. Classification for alzheimer's disease and frontotemporal dementia via resting-state electroencephalography-based coherence and convolutional neural network. *Cognitive Neurodynamics*, 19(1), 2025. ISSN 1871-4099. doi: 10.1007/s11571-025-10232-2. URL `http://dx.doi.org/10.1007/s11571-025-10232-2`.

[6] Dominik Klepl, Min Wu, and Fei He. Graph neural network-based eeg classification: A survey. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 32:493–503, 2024. ISSN

1558-0210. doi: 10.1109/tnsre.2024.3355750. URL http://dx.doi.org/10.1109/tnsre.2024.3355750.

[7] Dominik Klepl, Fei He, Min Wu, Daniel J. Blackburn, and Ptolemaios Sarrigiannis. Eeg-based graph neural network classification of alzheimer's disease: An empirical evaluation of functional connectivity methods. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 30:2651–2660, 2022. ISSN 1558-0210. doi: 10.1109/tnsre.2022.3204913. URL http://dx.doi.org/10.1109/tnsre.2022.3204913.

[8] Dominik Klepl, Fei He, Min Wu, Daniel J. Blackburn, and Ptolemaios Sarrigiannis. Adaptive gated graph convolutional network for explainable diagnosis of alzheimer's disease using eeg data. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 31:3978–3987, 2023. ISSN 1558-0210. doi: 10.1109/tnsre.2023.3321634. URL http://dx.doi.org/10.1109/tnsre.2023.3321634.

[9] Xiaocai Shan, Jun Cao, Shoudong Huo, Liangyu Chen, Ptolemaios Georgios Sarrigiannis, and Yifan Zhao. <scp>spatial–temporal</scp> graph convolutional network for alzheimer classification based on brain functional connectivity imaging of electroencephalogram. *Human Brain Mapping*, 43(17):5194–5209, 2022. ISSN 1097-0193. doi: 10.1002/hbm.25994. URL http://dx.doi.org/10.1002/hbm.25994.

[10] Qingjie Xu, Libing An, Haiqiang Yang, and Keum-Shik Hong. A multi-graph convolutional network method for alzheimer's disease diagnosis based on multi-frequency eeg data with dual-mode connectivity. *Frontiers in Neuroscience*, 19, 2025. ISSN 1662-453X. doi: 10.3389/fnins.2025.1555657. URL http://dx.doi.org/10.3389/fnins.2025.1555657.

[11] Ehssan Aljanabi and İlker Türker. Connectogram-coh: A coherence-based time-graph representation for eeg-based alzheimer's disease detection. *Diagnostics*, 15(11):1441, 2025. ISSN 2075-4418. doi: 10.3390/diagnostics15111441. URL http://dx.doi.org/10.3390/diagnostics15111441.

[12] Yujian Liu, Libing An, Haiqiang Yang, and Shuzhi Sam Ge. Multi-frequency eeg and multi-functional connectivity graph convolutional network based detection method of patients with alzheimer's disease. *Complex & Intelligent Systems*, 11(8), 2025. ISSN 2198-6053. doi: 10.1007/s40747-025-01974-x. URL http://dx.doi.org/10.1007/s40747-025-01974-x.

[13] Andreas Miltiadous, Katerina D. Tzimourta, Theodora Afrantou, Panagiotis Ioannidis, Nikolaos Grigoriadis, Dimitrios G. Tsalikakis, Pantelis Angelidis, Markos G. Tsipouras, Euripidis Glavas, Nikolaos Giannakeas, and Alexandros T. Tzallas. A dataset of scalp eeg recordings of alzheimer's disease, frontotemporal dementia and healthy subjects from routine eeg. *Data*, 8(6):95, 2023. ISSN 2306-5729. doi: 10.3390/data8060095. URL http://dx.doi.org/10.3390/data8060095.

[14] Andreas Miltiadous, Katerina D. Tzimourta, Theodora Afrantou, Panagiotis Ioannidis, Nikolaos Grigoriadis, Dimitrios G. Tsalikakis, Pantelis Angelidis, Markos G. Tsipouras, Evripidis Glavas, Nikolaos Giannakeas, and Alexandros T. Tzallas. A dataset of eeg recordings from: Alzheimer's disease, frontotemporal dementia and healthy subjects, 2024. URL https://openneuro.org/datasets/ds004504/versions/1.0.8.

[15] Andreas Miltiadous, Aimilia Ntetska, Vasileios Aspiotis, Efthalia Moustakli, Markos G. Tsipouras, Alexandros T. Tzallas, Nikolaos Giannakeas, Euripidis Glavas, Pantelis Angelidis, and Katerina D. Tzimourta. The ahepa eeg benchmark: setting the standard for machine learning in dementia diagnosis, a scoping review. *Cognitive Neurodynamics*, 20(1), 2026. ISSN 1871-4099. doi: 10.1007/s11571-026-10464-w. URL http://dx.doi.org/10.1007/s11571-026-10464-w.

[16] Sayash Kapoor and Arvind Narayanan. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 4(9):100804, 2023. ISSN 2666-3899. doi: 10.1016/j.patter.2023.100804. URL http://dx.doi.org/10.1016/j.patter.2023.100804.

[17] Matthew Rosenblatt, Link Tejavibulya, Rongtao Jiang, Stephanie Noble, and Dustin Scheinost. Data leakage inflates prediction performance in connectome-based machine learning models. *Nature Communications*, 15(1):1829, 2024. doi: 10.1038/s41467-024-46150-w. URL https://doi.org/10.1038/s41467-024-46150-w.

[18] Sohrab Saeb, Luca Lonini, Arun Jayaraman, David C. Mohr, and Konrad P. Kording. The need to approximate the use-case in clinical machine learning. *GigaScience*, 6(5):gix019, 2017. doi: 10.1093/gigascience/gix019. URL https://doi.org/10.1093/gigascience/gix019.

[19] Geoffrey Brookshire, Jake Kasper, Nicholas M. Blauch, Yunan Charles Wu, Ryan Glatt, David A. Merrill, Spencer Gerrol, Keith J. Yoder, Colin Quirk, and Ché Lucero. Data leakage in deep learning studies of translational eeg. *Frontiers in Neuroscience*, 18:1373515, 2024. doi: 10.3389/fnins.2024.1373515. URL https://doi.org/10.3389/fnins.2024.1373515.

[20] Burak Aggul. Window leakage in clinical eeg machine learning: matched subject-independent evaluation across adhd and parkinson cohorts. *Frontiers in Neuroinformatics*, 20:1930795, 2026. doi: 10.3389/fninf.2026.1930795. URL https://doi.org/10.3389/fninf.2026.1930795.

[21] Abdulnasır Yıldız and Hasan Zan. Deep learning-based alzheimer's disease detection from multi-channel eeg using fused time–frequency image grids. *Diagnostics*, 16(5):746, 2026. ISSN 2075-4418. doi: 10.3390/diagnostics16050746. URL http://dx.doi.org/10.3390/diagnostics16050746.

[22] Thawirasm Jungrungrueang, Sawrawit Chairat, Kasidach Rasitanon, Praopim Limsakul, and Krit Charupanit. Translational approach for dementia subtype classification using convolutional neural network based on eeg connectome dynamics. *Scientific Reports*, 15(1), 2025. ISSN 2045-2322. doi: 10.1038/s41598-025-02018-7. URL http://dx.doi.org/10.1038/s41598-025-02018-7.

[23] Shuzhi Zhao, Guangyan Dai, Jingting Li, Xiaoxia Zhu, Xiyan Huang, Yongxue Li, Mingdan Tan, Lan Wang, Peng Fang, Xi Chen, Nan Yan, and Hanjun Liu. An interpretable model based on graph learning for diagnosis of parkinson's disease with voice-related eeg. *npj Digital Medicine*, 7(1), 2024. ISSN 2398-6352. doi: 10.1038/s41746-023-00983-9. URL http://dx.doi.org/10.1038/s41746-023-00983-9.

[24] Daniel A. Nation, Melanie D. Sweeney, Axel Montagne, Abhay P. Sagare, Lina M. D'Orazio, Maricarmen Pachicano, Farshid Sepehrband, Amy R. Nelson, David P. Buennagel, Michael G. Harrington, Tammie L. S. Benzinger, Anne M. Fagan, John M. Ringman, Lon S. Schneider, John C. Morris, Helena C. Chui, Meng Law, Arthur W. Toga, and Berislav V. Zlokovic. Blood–brain barrier breakdown is an early biomarker of human cognitive dysfunction. *Nature Medicine*, 25(2):270–276, 2019. ISSN 1546-170X. doi: 10.1038/s41591-018-0297-y. URL http://dx.doi.org/10.1038/s41591-018-0297-y.

[25] Melanie D. Sweeney, Abhay P. Sagare, and Berislav V. Zlokovic. Blood–brain barrier breakdown in alzheimer disease and other neurodegenerative disorders. *Nature Reviews Neurology*, 14(3):133–150, 2018. ISSN 1759-4766. doi: 10.1038/nrneurol.2017.188. URL http://dx.doi.org/10.1038/nrneurol.2017.188.

[26] Dan Z. Milikovsky, Jonathan Ofer, Vladimir V. Senatorov, Aaron R. Friedman, Ofer Prager, Liron Sheintuch, Netta Elazari, Ronel Veksler, Daniel Zelig, Itai Weissberg, Guy Bar-Klein, Evyatar Swissa, Erez Hanael, Gal Ben-Arie, Osnat Schefenbauer, Lyna Kamintsky, Rotem Saar-Ashkenazy, Ilan Shelef, Merav H. Shamir, Ilan Goldberg, Amir Glik, Felix Benninger, Daniela Kaufer, and Alon Friedman. Paroxysmal slow cortical activity in alzheimer's disease and epilepsy is associated with blood-brain barrier dysfunction. *Science Translational Medicine*, 11(521), 2019. ISSN 1946-6242. doi: 10.1126/scitranslmed.aaw8954. URL http://dx.doi.org/10.1126/scitranslmed.aaw8954.

[27] Vladimir V. Senatorov, Aaron R. Friedman, Dan Z. Milikovsky, Jonathan Ofer, Rotem Saar-Ashkenazy, Adiel Charbash, Naznin Jahan, Gregory Chin, Eszter Mihaly, Jessica M. Lin, Harrison J. Ramsay, Ariana Moghbel, Marcela K. Preininger, Chelsy R. Eddings, Helen V. Harrison, Rishi Patel, Yishuo Shen, Hana Ghanim, Huanjie Sheng, Ronel Veksler, Peter H. Sudmant, Albert Becker, Barry Hart, Michael A. Rogawski, Andrew Dillin, Alon Friedman, and Daniela Kaufer. Blood-brain barrier dysfunction in aging induces hyperactivation of tgfβ signaling and chronic yet reversible neural dysfunction. *Science Translational Medicine*, 11 (521), 2019. ISSN 1946-6242. doi: 10.1126/scitranslmed.aaw8283. URL http://dx.doi.org/10.1126/scitranslmed.aaw8283.

[28] Dan Z. Milikovsky, Yotam Sharabi, Nir Giladi, Anat Mirelman, Ronen Sosnik, Firas Fahoum, and Inbal Maidan. Paroxysmal slow-wave events are uncommon in parkinson's disease. *Sensors*, 23(2):918, 2023. ISSN 1424-8220. doi: 10.3390/s23020918. URL http://dx.doi.org/10.3390/s23020918.

[29] S. Amara Ganon, A. Friedman, and Y. Zigel. Electroencephalogram data-based analysis of paroxysmal slow wave events patterns in brain pathologies. In *2025 47th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, page 1–7. IEEE, 2025. doi: 10.1109/embc58623.2025.11254025. URL http://dx.doi.org/10.1109/embc58623.2025.11254025.

[30] Thomas N. Kipf and Max Welling. Semi-supervised classification with graph convolutional networks, 2016. URL https://arxiv.org/abs/1609.02907.

[31] Shahid Iqbal, Humaira Nisar, and Kim Ho Yeap. Quantitative eeg signatures of power and functional connectivity alterations in alzheimer's disease and frontotemporal dementia. *Scientific Reports*, 16(1), 2026. ISSN 2045-2322. doi: 10.1038/s41598-026-42452-9. URL http://dx.doi.org/10.1038/s41598-026-42452-9.

[32] Kassra Ghassemkhani, Kevin S. Saroka, and Blake T. Dotta. Evaluating eeg complexity and spectral signatures in alzheimer's disease and frontotemporal dementia: evidence for rostrocaudal asymmetry. *npj Aging*, 11(1), 2025. ISSN 2731-6068. doi: 10.1038/s41514-025-00243-y. URL http://dx.doi.org/10.1038/s41514-025-00243-y.

[33] Mingkan Shen, Peng Wen, Bo Song, and Yan Li. Muti-band morlet mutual information functional connectivity for classifying alzheimer's disease and frontotemporal dementia with a deep learning technique. *Computers in Biology and Medicine*, 197:111041, 2025. ISSN 0010-4825. doi: 10.1016/j.compbiomed.2025.111041. URL http://dx.doi.org/10.1016/j.compbiomed.2025.111041.

[34] Xin Zhang, Ye Li, Fei Pi, and Xiu Zhang. Rgb-style input representations for eeg: Evaluating spatial concatenation versus band-wise stacking in deep emotion recognition. *Brain Sciences*,

16(7):716, 2026. ISSN 2076-3425. doi: 10.3390/brainsci16070716. URL http://dx.doi.org/10.3390/brainsci16070716.

[35] Petar Veličković, Guillem Cucurull, Arantxa Casanova, Adriana Romero, Pietro Liò, and Yoshua Bengio. Graph attention networks, 2017. URL https://arxiv.org/abs/1710.10903.

[36] Mohamed Radwan, Pedro G. Lind, and Anis Yazidi. Frequency bands eeg biomarkers for dementia using graph neural networks. bioRxiv preprint (not peer reviewed), 2025. URL http://dx.doi.org/10.1101/2025.08.18.670945.

[37] Cornelis J. Stam, Guido Nolte, and Andreas Daffertshofer. Phase lag index: Assessment of functional connectivity from multi channel eeg and meg with diminished bias from common sources. *Human Brain Mapping*, 28(11):1178–1193, 2007. ISSN 1097-0193. doi: 10.1002/hbm.20346. URL http://dx.doi.org/10.1002/hbm.20346.

[38] Martin Vinck, Robert Oostenveld, Marijn van Wingerden, Franscesco Battaglia, and Cyriel M.A. Pennartz. An improved index of phase-synchronization for electrophysiological data in the presence of volume-conduction, noise and sample-size bias. *NeuroImage*, 55(4):1548–1565, 2011. ISSN 1053-8119. doi: 10.1016/j.neuroimage.2011.01.055. URL http://dx.doi.org/10.1016/j.neuroimage.2011.01.055.

[39] Jiayi Cao, Bin Li, and Xiaoou Li. Identification of alzheimer's disease brain networks based on eeg phase synchronization. *BioMedical Engineering OnLine*, 24(1), 2025. ISSN 1475-925X. doi: 10.1186/s12938-025-01361-0. URL http://dx.doi.org/10.1186/s12938-025-01361-0.

[40] Alexander Kamrud, Brett Borghetti, and Christine Schubert Kabban. The effects of individual differences, non-stationarity, and the importance of data partitioning decisions for training and testing of eeg cross-participant models. *Sensors*, 21(9):3225, 2021. doi: 10.3390/s21093225. URL https://doi.org/10.3390/s21093225.

[41] Russell A. Poldrack, Grace Huckins, and Gael Varoquaux. Establishment of best practices for evidence for prediction. *JAMA Psychiatry*, 77(5):534, 2020. doi: 10.1001/jamapsychiatry.2019.3671. URL https://doi.org/10.1001/jamapsychiatry.2019.3671.

[42] Gaël Varoquaux. Cross-validation failure: Small sample sizes lead to large error bars. *NeuroImage*, 180:68–77, 2018. doi: 10.1016/j.neuroimage.2017.06.061. URL https://doi.org/10.1016/j.neuroimage.2017.06.061.

[43] Alexandre Gramfort. Meg and eeg data analysis with mne-python. *Frontiers in Neuroscience*, 7, 2013. ISSN 1662-453X. doi: 10.3389/fnins.2013.00267. URL http://dx.doi.org/10.3389/fnins.2013.00267.

[44] Te-Won Lee, Mark Girolami, and Terrence J. Sejnowski. Independent component analysis using an extended infomax algorithm for mixed subgaussian and supergaussian sources. *Neural Computation*, 11(2):417–441, 1999. ISSN 1530-888X. doi: 10.1162/089976699300016719. URL http://dx.doi.org/10.1162/089976699300016719.

[45] Luca Pion-Tonachini, Ken Kreutz-Delgado, and Scott Makeig. Iclabel: An automated electroencephalographic independent component classifier, dataset, and website. *NeuroImage*, 198:181–197, 2019. ISSN 1053-8119. doi: 10.1016/j.neuroimage.2019.05.026. URL http://dx.doi.org/10.1016/j.neuroimage.2019.05.026.

[46] P. Welch. The use of fast fourier transform for the estimation of power spectra: A method based on time averaging over short, modified periodograms. *IEEE Transactions on Audio and Electroacoustics*, 15(2):70–73, 1967. ISSN 0018-9278. doi: 10.1109/tau.1967.1161901. URL http://dx.doi.org/10.1109/TAU.1967.1161901.

[47] Gregory Lee, Ralf Gommers, Filip Waselewski, Kai Wohlfahrt, and Aaron O'Leary. Pywavelets: A python package for wavelet analysis. *Journal of Open Source Software*, 4(36):1237, 2019. ISSN 2475-9066. doi: 10.21105/joss.01237. URL http://dx.doi.org/10.21105/joss.01237.

[48] Bo Hjorth. Eeg analysis based on time domain properties. *Electroencephalography and Clinical Neurophysiology*, 29(3):306–310, 1970. ISSN 0013-4694. doi: 10.1016/0013-4694(70)90143-4. URL http://dx.doi.org/10.1016/0013-4694(70)90143-4.

[49] Joshua S. Richman and J. Randall Moorman. Physiological time-series analysis using approximate entropy and sample entropy. *American Journal of Physiology-Heart and Circulatory Physiology*, 278(6):H2039–H2049, 2000. ISSN 1522-1539. doi: 10.1152/ajpheart.2000.278.6.h2039. URL http://dx.doi.org/10.1152/ajpheart.2000.278.6.H2039.

[50] Jukka-Pekka Onnela, Jari Saramäki, János Kertész, and Kimmo Kaski. Intensity and coherence of motifs in weighted complex networks. *Physical Review E*, 71(6), 2005. ISSN 1550-2376. doi: 10.1103/physreve.71.065103. URL http://dx.doi.org/10.1103/PhysRevE.71.065103.

[51] Vernon J Lawhern, Amelia J Solon, Nicholas R Waytowich, Stephen M Gordon, Chou P Hung, and Brent J Lance. Eegnet: a compact convolutional neural network for eeg-based brain–computer interfaces. *Journal of Neural Engineering*, 15(5):056013, 2018. ISSN 1741-2552. doi: 10.1088/1741-2552/aace8c. URL http://dx.doi.org/10.1088/1741-2552/aace8c.

[52] Arun Singh, Rachel Cole, Arturo Espinoza, Jim Cavanagh, and Nandakumar Narayanan. Rest eyes open (openneuro accession ds004584, version 1.0.0), 2023. URL https://openneuro.org/datasets/ds004584/versions/1.0.0.

[53] Claude Nadeau and Yoshua Bengio. Inference for the generalization error. *Machine Learning*, 52(3):239–281, 2003. ISSN 1573-0565. doi: 10.1023/a:1024068626366. URL http://dx.doi.org/10.1023/A:1024068626366.

[54] Sture Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70, 1979.

[55] Janez Demšar. Statistical comparisons of classifiers over multiple data sets. *Journal of Machine Learning Research*, 7:1–30, 2006. URL https://jmlr.org/papers/v7/demsar06a.html.

Supplementary material

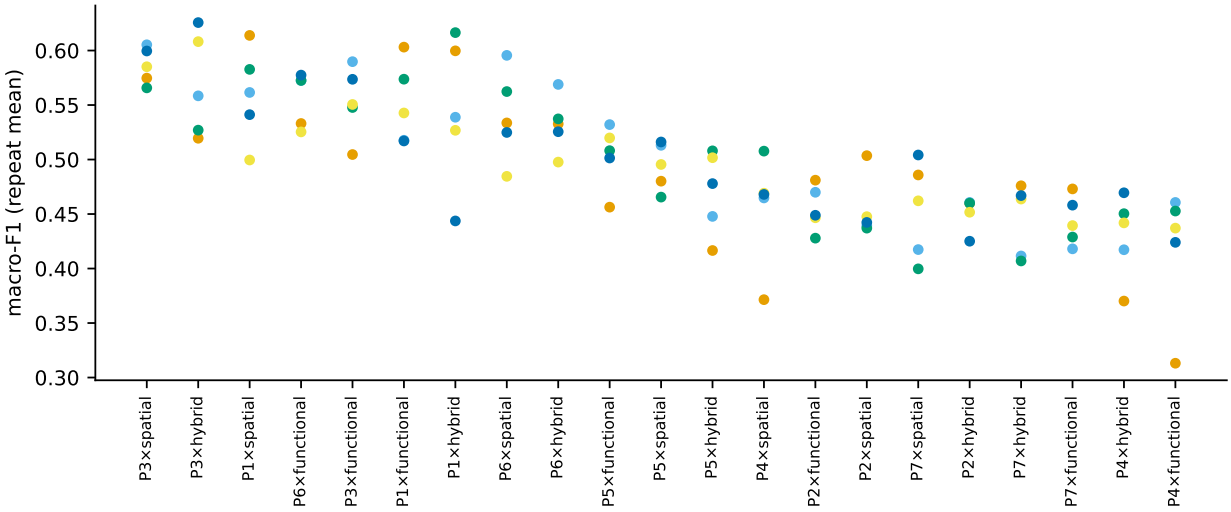


Figure S1: Mean macro-F1 of each repeat (seed) for every cell on the selection folds: variation due to the cross-validation partition and initialisation.

Table S1: Paired comparison of each representation's best GCN cell with its best non-graph twin under both training regimes (25 selection splits). Δ: GCN minus classical; p: Holm-corrected Wilcoxon and Nadeau–Bengio tests within each regime.

Regime	Representation	GCN	Classical	GCN F1	Classical F1	Δ	p_{W}	p_{NB}
train only	P1 spectral	GCN (spatial)	Random forest	0.560	0.541	+0.019	1.00	1.00
train only	P2 connectivity	GCN (functional)	RBF-SVM	0.455	0.459	-0.004	1.00	1.00
train only	P3 wavelet	GCN (spatial)	Random forest	0.586	0.536	+0.050	0.40	1.00
train only	P4 time domain	GCN (spatial)	Random forest	0.456	0.481	-0.025	1.00	1.00
train only	P6 fused	GCN (functional)	RBF-SVM	0.556	0.561	-0.004	1.00	1.00
train only	P7 PSWE	GCN (spatial)	Random forest	0.454	0.498	-0.044	0.31	1.00
train + val	P1 spectral	GCN (functional)	Random forest	0.558	0.588	-0.030	0.92	1.00
train + val	P2 connectivity	GCN (spatial)	RBF-SVM	0.453	0.435	+0.019	1.00	1.00
train + val	P3 wavelet	GCN (spatial)	Random forest	0.613	0.581	+0.033	0.80	1.00
train + val	P4 time domain	GCN (spatial)	Random forest	0.439	0.487	-0.049	0.38	1.00
train + val	P6 fused	GCN (spatial)	RBF-SVM	0.580	0.579	+0.000	1.00	1.00
train + val	P7 PSWE	GCN (spatial)	Random forest	0.456	0.502	-0.046	0.24	1.00

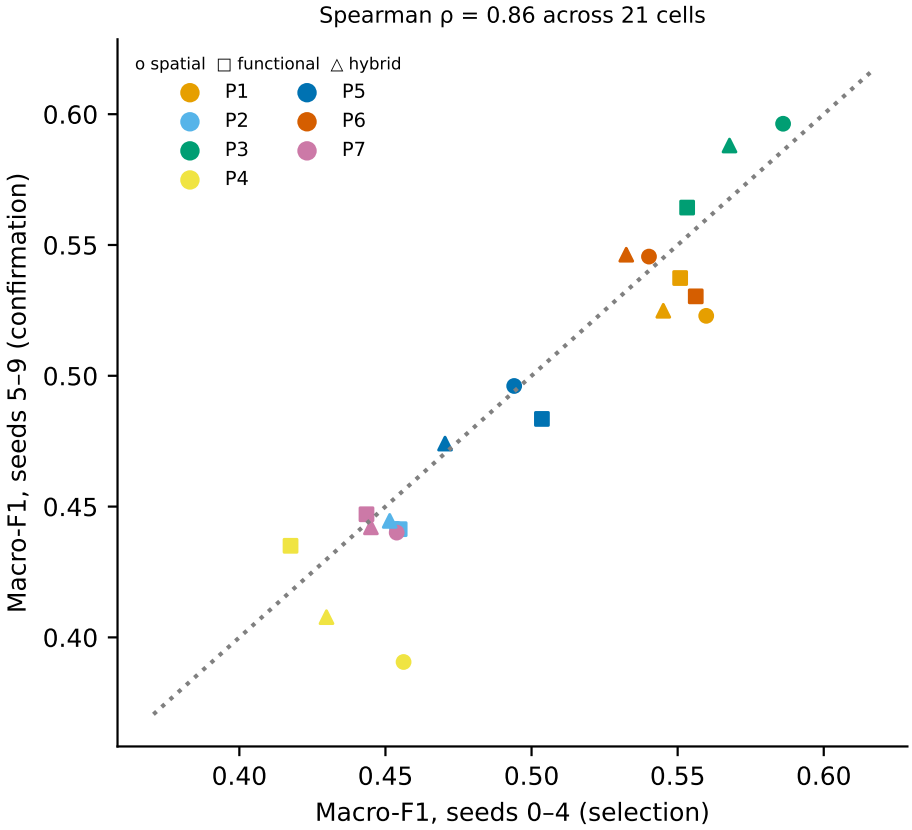


Figure S2: Every cell's mean macro-F1 on the selection folds (seeds 0–4) against the independent confirmation folds (seeds 5–9); dashed line, identity. Rank correlation: Spearman $\rho = 0.86$.

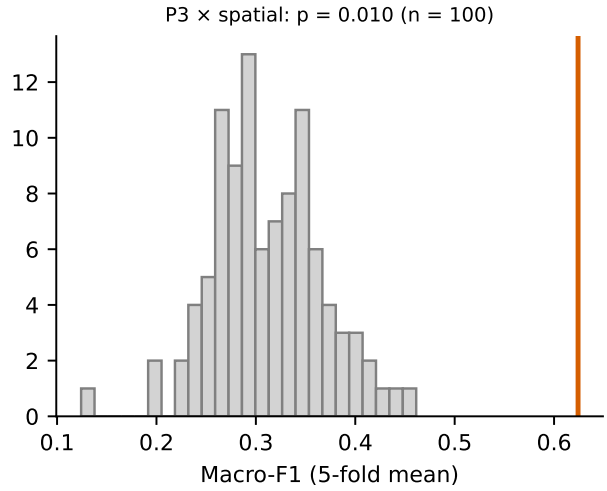


Figure S3: Label-permutation null distribution (100 permutations, first confirmation repeat's five folds) for the best cell; red line, observed macro-F1 on the same splits.

Table S2: Ablations on the selection folds (seeds 0–4), on which the two reference cells were chosen; layout as Table 8. These estimates are subject to selection bias and are given for comparison only.

	P3 × spatial		P3 × hybrid	
Ablation	Δ [95% CI]	p_{Holm}	Δ [95% CI]	p_{Holm}
Age/sex regressed out	-0.046 [-0.089, -0.003]	1.00	-0.044 [-0.093, +0.004]	1.00
No graph (identity)	-0.043 [-0.088, +0.003]	1.00	-0.024 [-0.064, +0.015]	1.00
Random graph (degree-preserving)	-0.037 [-0.075, +0.002]	1.00	-0.018 [-0.049, +0.013]	1.00
Fully connected	-0.066 [-0.113, -0.020]	1.00	-0.032 [-0.068, +0.005]	1.00
$k = 3$	-0.022 [-0.047, +0.004]	1.00	-0.009 [-0.026, +0.007]	1.00
$k = 6$	-0.009 [-0.033, +0.015]	1.00	+0.000 [-0.031, +0.032]	1.00
Binarised edges	-0.006 [-0.020, +0.008]	1.00	-0.023 [-0.057, +0.011]	1.00
Depth 1	-0.026 [-0.073, +0.020]	1.00	-0.006 [-0.042, +0.031]	1.00
Depth 3	-0.010 [-0.042, +0.022]	1.00	+0.007 [-0.018, +0.033]	1.00
GAT instead of GCN	-0.022 [-0.069, +0.025]	1.00	-0.021 [-0.044, +0.002]	1.00
No subject-equal weighting	+0.007 [-0.030, +0.044]	1.00	+0.014 [-0.016, +0.044]	1.00
+ P7 PSWE features	-0.024 [-0.059, +0.011]	1.00	+0.005 [-0.028, +0.038]	1.00
PSWE co-occurrence edges	-0.029 [-0.065, +0.007]	1.00	-0.011 [-0.044, +0.022]	1.00

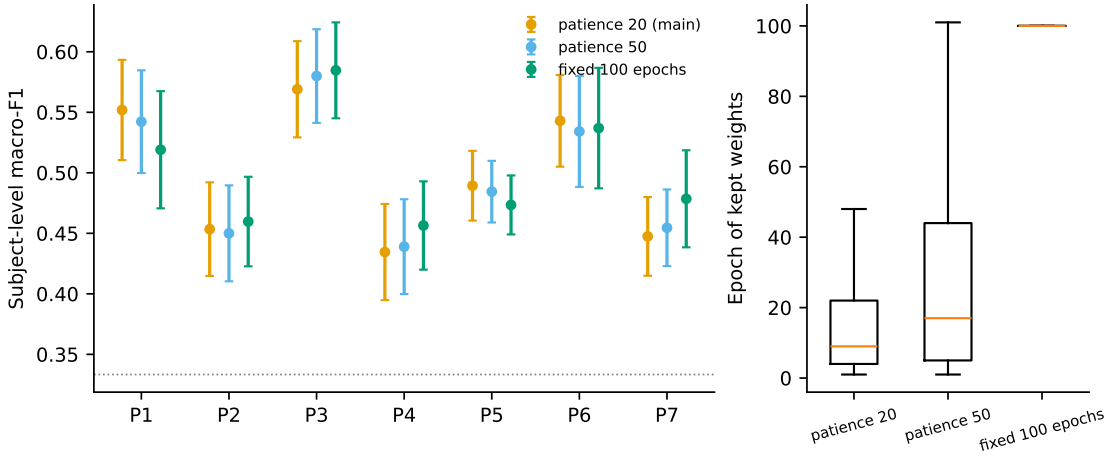


Figure S4: Stopping-rule sensitivity: macro-F1 per representation (mean and 95% CI over the 25 selection splits, averaged over graph types) under each stopping rule (left) and the epoch of the kept weights (right).

Table S3: Per-representation macro-F1 under each stopping rule (selection splits, averaged over graph types): change from the main grid and Holm-corrected Wilcoxon p (paired over the 25 splits).

	Patience 20	Patience 50			Fixed 100 epochs		
Representation	Macro-F1	Macro-F1	Δ	p_{Holm}	Macro-F1	Δ	p_{Holm}
P1	0.552	0.542	-0.010	0.56	0.519	-0.033	0.31
P2	0.453	0.450	-0.003	1.00	0.460	+0.006	1.00
P3	0.569	0.580	+0.011	0.97	0.585	+0.016	0.75
P4	0.435	0.439	+0.004	0.97	0.456	+0.022	0.35
P5	0.489	0.484	-0.005	1.00	0.473	-0.016	0.75
P6	0.543	0.534	-0.009	1.00	0.537	-0.006	1.00
P7	0.447	0.455	+0.007	1.00	0.479	+0.031	0.18

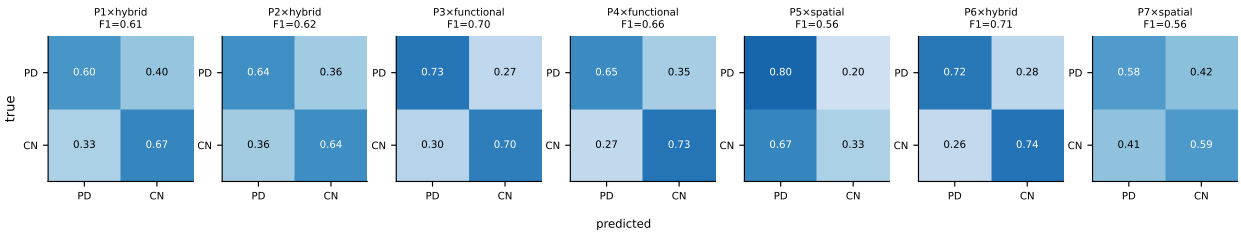


Figure S5: External cohort ds004584: row-normalised confusion matrices (summed over the 25 test splits) of the best graph type for each representation.

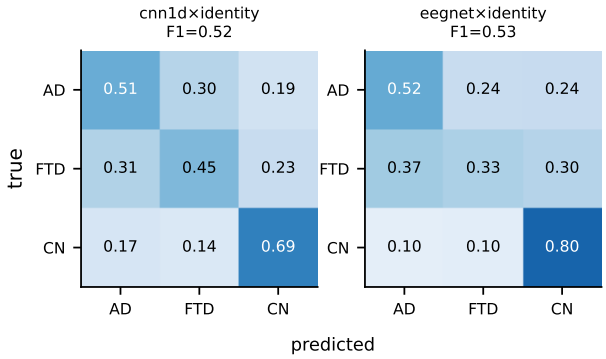


Figure S6: Deep-learning baselines on raw windows (ds004504, 25 selection splits): row-normalised confusion matrices of EEGNet and the 1D-CNN.